DRIVING BEHAVIOUR ANALYSIS USING DEEP LEARNING WITH DATA AUGMENTATION TECHNIQUE

AMIR IZHAN BIN BEKRI 2121507

PROGRAMME:

BACHELOR OF ELECTRICAL AND ELECTRONIC ENGINEERING

FINAL YEAR PROJECT 2

SUPERVISOR: Assoc. Prof. Dr. Malik Arman Bin Morshidi

Department of Electrical and Computer Engineering

KULLIYYAH OF ENGINEERING

INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA

JANUARY 2026

DECLARATION

I hereby declare that this is the result of my own investigations, except where otherwise stated. I also declare that it has not been previously or concurrently submitted for any other degrees at IIUM (International Islamic University Malaysia) or other institutions.

( )

.Amir Izhan Bin Bekri..... Date: 16 JANUARY 2026

APPROVAL PAGE

I certify that I have supervised and read this study and that, it conforms to acceptable standards of scholarly presentation and is fully adequate, in scope and quality, as Final Year Project report as partial fulfilment for a degree of Bachelor of Electrical and Electronic Engineering (Honours).

.....

[Assoc.Prof.Dr. Malik Arman Bin Morshidi]
Supervisor

ABSTRACT

Unsafe driving behaviours such as aggressive and drowsy driving remain major contributors to road accidents. Many existing driving behaviour analysis systems rely on basic indicators, fail to capture the temporal nature of driving data, and are affected by class imbalance, leading to unreliable predictions.

This project proposes a hybrid deep learning framework for driving behaviour analysis using time-series sensor data. The approach combines Conv1D and Bidirectional Long Short-Term Memory (BiLSTM) networks to model both short-term motion patterns and long-term temporal dependencies. Accelerometer and GPS data from the public UAH-DriveSet are preprocessed, synchronised, and segmented into fixed-length windows. A dual-input feature design using sequential and statistical features is adopted to enhance behaviour representation.

To address data imbalance, Conditional Tabular Generative Adversarial Networks (CTGAN) are used to generate synthetic samples for minority driving behaviours. The model is evaluated at the window level using standard classification metrics. Results show improved performance compared to baseline methods. Future work will integrate contextual information such as road type, weather, and time of day to improve robustness.

ACKNOWLEDGEMENTS

First and foremost, I would like to express my deepest gratitude to my supervisor, Assoc. Prof. Dr. Malik Arman, for the highest regard in accepting me under his supervision and providing me with the opportunity to explore this fascinating topic. I am truly grateful for his professional advice, valuable suggestions, patience, and continuous encouragement throughout my FYP process. His encouragement and trust in my ability have been a significant motivational stimulus, pushing me to give my best effort.

In addition, I also owe the deepest debt of gratitude to my parents, whose unwavering love, sacrifices, and constant prayers have been the foundation of my success. Their support has allowed me to reach this level of my life, and their trust in me has been a principal motive behind my determination to complete my degree. My ultimate goal is to make them proud and provide for them in the years to come, repaying them for the love and care they have shown me.

I would like to acknowledge the International Islamic University Malaysia (IIUM) for having endowed me with the resources available, facilities, and learning environment that highly supported my research. The assistance from the university has been instrumental in enabling me to realize my academic goals.

I would also like to express my appreciation to the FYP coordinators as well as all lecturers who have their knowledge, guidelines, and words of advice, and have been very helpful to me throughout this semester.

Lastly, I would like to thank my friends, course mates, and roommates. They have been the pillars of my support team during my time in campus, offering emotional support, helpful advice, and helping hands in all ways possible. Their presence has been a great source of strength and encouragement throughout this journey.

TABLE OF CONTENTS

ABSTRACT iv

ACKNOWLEDGEMENTS v

TABLE OF CONTENTS vi

LIST OF TABLES xi

LIST OF FIGURES xii

LIST OF ABBREVIATIONS xiv

CHAPTER 1 1

INTRODUCTION 1

1.1 OVERVIEW 1

1.2 PROBLEM STATEMENT 2

1.3 OBJECTIVE 3

1.4 METHODOLOGY 4

1.5 SCOPE OF STUDY 5

1.6 REPORT ORGANIZATION 5

CHAPTER 2 6

2.1 OVERVIEW 6

2.2 DRIVING BEHAVIOUR AND UNSAFE DRIVING PATTERNS 7

2.3 DRIVING DATA SOURCES AND FEATURE REPRESENTATION 9

2.3.1 Driving Data Sources 9

2.3.2 Feature Representation Techniques 10

2.3.3 Discussion and Research Implications 11

2.4 KEY DRIVING BEHAVIOUR PARAMETERS 12

2.4.1 Primary Driving Parameters 12

2.4.2 Contextual Behaviour Conditions 13

2.5 TRADITIONAL MACHINE LEARNING APPROACHES FOR DRIVING BEHAVIOUR ANALYSIS 15

2.5.1 Discussion 17

2.6 DEEP LEARNING METHODS FOR TIME SERIES DRIVING BEHAVIOUR ANALYSIS 18

2.6.1 CNN and LSTM Behaviour Recognition 18

2.6.2 Hybrid and Generative Approaches 20

2.6.3 Discussion 22

2.7 TRANSFER LEARNING AND OBJECT DETECTION 23

2.8 PREVIOUS WORK ON DRIVING BEHAVIOUR ANALYSIS 26

2.9 SUMMARY 29

CHAPTER 3 31

3.1 OVERVIEW 31

3.2 PROCESS FLOWCHART 32

3.3 DATASET 34

3.3.1 Data Source and Description 35

3.3.2 Data Preprocessing and Cleaning 36

3.3.3 Window Segmentation Strategy 38

3.3.4 Feature Construction 40

3.4 PROPOSED MODEL ARCHITECTURE 42

3.4.1 Overview of the Hybrid Architecture 43

3.4.2 Sequence Branch (Conv1D + BiLSTM) 44

3.4.3 Statistical Feature Branch 46

3.4.4 Feature Fusion and Classification Head 47

3.4.5 Loss Function 48

3.5 DATA AUGMENTATION STRATEGY 49

3.6 MODEL TRAINING AND EVALUATION SETUP 51

3.6.1 Training Configuration 51

3.6.2 Window-Level Evaluation 52

3.6.3 Evaluation Metrics 52

3.7 SYSTEM FLOWCHART 53

3.8 DEVELOPMENT TOOLS AND ENVIRONMENT 55

3.8.1 Programming Languages and Libraries 55

3.9 SUMMARY 56

CHAPTER 4 58

4.1 OVERVIEW 58

4.2 EXPERIMENTAL SETUP 58

4.2.1 Dataset Partitioning 58

4.2.2 Window Segmentation 59

4.2.3 Input Features 60

4.2.4 Model Training Configuration 61

4.3 BASELINE MODEL PERFORMANCE (WITHOUT DATA AUGMENTATION) 62

4.3.1 Window-Level Classification Performance 62

4.3.2 Confusion Matrix Analysis Baseline Model 64

4.4 IMPACT OF DATA AUGMENTATION ON MODEL PERFORMANCE 66

4.4.1 Performance After Data Augmentation 66

4.4.2 Confusion Matrix Analysis After Data Augmentation 68

4.4.3 Training Behaviour Across Folds 69

4.4.4 Comparative Analysis 73

4.5 COMPARISON WITH EXISTING METHODS 75

4.5.1 Linear Support Vector Machine (SVM) 75

4.5.2 Adaptive Boosting (AdaBoost) Classifier 77

4.5.3 Random Forest Classifier 79

4.5.4 Gradient Boosting Classifier 80

4.5.5 K-Nearest Neighbours (KNN) Classifier 82

4.5.6 CTGAN-Enhanced BiLSTM with Hoeffding Tree Classifier 84

4.5.7 Long Short-Term Memory (LSTM) Architecture 86

4.5.8 Convolutional Neural Network with Long Short-Term Memory (CNN–LSTM) Architecture 88

4.5.9 Summary 90

4.6 KEY FINDINGS 91

4.7 LIMITATIONS 94

CHAPTER 5 96

5.1 CONCLUSION 96

5.2 FUTURE WORK 97

BIBLIOGRAPHY 99

LIST OF TABLES

Table 2.1 Summary of Driving Data Sources, Feature Representation, and Advantages 11

Table 2.2: Traditional Machine Learning Approaches for Driving Behaviour Analysis 17

Table 2.3: Deep Learning Approaches for Driving Behaviour Analysis.....21

Table 2.4 Model Evaluation Metrics for YOLOv7, YOLOv8, and YOLOv9 [22].....24

Table 2.5: Benchmark Table For Driving Behaviour Analysis Using ML.....28

Table 3.1 List of driver and vehicles that performed the tests [23].....35

Table 3.2 Distribution of Driving Behaviour Classes in the UAH-DriveSet [23].....36

Table 4.1 Baseline Model CNN-BILSTM Classification Report.....63

Table 4.2 Augmented Baseline Model Classification Report.....67

Table 4.3: SVM Classification Report.....76

Table 4.4: AdaBoost Classification Report 78

Table 4.5: Random Forest Classification Report.....79

Table 4.6: Gradient Boosting Classification Report.....81

Table 4.7: KNN Classification Report.....83

Table 4.8:CTGAN-Enhanced BiLSTM with Hoeffding Tree Classifier Classification Report.....85

Table 4.9: LSTM Classification Report.....87

Table 4.10: CNN LSTM Classification Report 89

Table 4.11: Performance Comparison of Existing and Proposed Models at Window Level 93

LIST OF FIGURES

Figure 1.1 Malaysia Road Accidents and Fatalities 2010 – 2021 1

Figure 2.1 Driver-based and vehicle-based features used for fatigue and drowsiness detection [5] 7

Figure 2.2 Driver maneuvers and vehicle state features used for aggressive driving detection [5] 8

Figure 2.3: Layered Driver and Environmental Context Architecture in SafeSmartDrive [22] 14

Figure 2.4 Driving behaviour classification BiLSTM vs. SLSTM [12] 19

Figure 2.5 Data augmentation process CTGAN [12] 22

Figure 2.6: YOLOv9 Confusion Matrix [22] 24

Figure 2.7: YOLOv7 and YOLOv8 Confusion Matrices [22] 25

Figure 2.8: SafeSmartDrive System Architecture [22] 25

Figure 3.1 Project Flowchart Methodology 33

Figure 3.2: Sample of preprocessed Driving Behaviour Dataset 38

Figure 3.3 Block diagram of the proposed hybrid Conv1D–BiLSTM and statistical feature fusion architecture 44

Figure 3.4: CTGAN-based data augmentation framework for generating synthetic driving behaviour samples [12] 49

Figure 3.5: Sample Synthetic Aggressive Behaviour Driving Data Generated Using CTGAN 50

Figure 3.6: Sample Synthetic Drowsy Behaviour Driving Data Generated Using CTGAN 50

Figure 3.7 Flowchart propose solution 54

Figure 4.1: Confusion Matrix Analysis Baseline Model 65

Figure 4.2: 1.1.1 Confusion Matrix Analysis After Data Augmentation.....69

Figure 4.3: Model Accuracy Fold 1 71

Figure 4.4: Model Accuracy Fold 2 72

Figure 4.5: Model Accuracy Fold 3 72

Figure 4.6: Model Accuracy Fold 4 72

Figure 4.7: Model Accuracy Fold 5 73

Figure 4.8 SVM Confusion Matrix 76

Figure 4.9: AdaBoost Confusion Matrix 78

Figure 4.10: Random Forest Confusion Matrix 80

Figure 4.11: Gradient Boosting Confusion Matrix 82

Figure 4.12: KNN Confusion Matrix 84

Figure 4.13: CTGAN-Enhanced BiLSTM with Hoeffding Tree Classifier Confusion MATRIX 86

Figure 4.14: LSTM Confusion Matrix 88

Figure 4.15: CNN LSTM Confusion Matrix 90

LIST OF ABBREVIATIONS

1. ML = Machine Learning
2. DL = Deep Learning
3. OBD = On-Board Diagnostics
4. SVM = Support Vector Machine
5. CNN = Convolutional Neural Network
6. LSTM = Long Short-Term Memory
7. KNN = K-Nearest Neighbors
8. ResNet = Residual Network
9. mAp = Mean Average Precision
10. YOLO = You Only Look Once
11. RF = Random Forest
12. AUC = Area Under the Curve
13. MLP = Multi-Layer Perceptron
14. ROC = Receiver Operating Characteristic
15. BiLSTM = Bidirectional Long Short-Term Memory
16. SLSTM = Stacked Long Short-Term Memory
17. CTGAN = Conditional Tabular Generative Adversarial Network
18. MS COCO = Microsoft Common Objects in Context
19. DPN = Dual Path Network
20. i-DREAMS = Intelligent Driver and Road Environment Assessment and Monitoring System
21. SHRP2 = Second Strategic Highway Research Program
22. EEG = Electroencephalogram
23. ECG = Electrocardiogram
24. EMG = Electromyogram
25. GSR = Galvanic Skin Response
26. MMPNet = Multimodal Physiological Network
27. LDA = Linear Discriminant Analysis
28. PAA = Piecewise Aggregate Approximation
29. DTW = Dynamic Time Warping
30. SAX = Symbolic Aggregate approximation
31. ILUBISTAR = Intelligent Learning Using Biometric-Informed Spatiotemporal Analysis for Risk
32. UBI = Usage-Based Insurance
33. ITS = Intelligent Transportation Systems

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Road accidents are one of the main causes of death around the world. The World Health Organization reports that over 1.3 million people die every year in automobile accidents. The majority of these collisions are caused by risky driving behaviors like drowsiness, distraction, or aggressive driving [18]. These behaviors reduce a driver's ability to remain alert and make safe choices on the road. In Malaysia, road accidents and fatalities were high from 2010 to 2021, as illustrated by Figure 1.1 [18].

Figure 1.1 Malaysia Road Accidents and Fatalities 2010 – 2021

However, to solve this problem, there has been a growing trend towards the application of intelligent and automated systems for the analysis of driving behavior and the prevention of accidents. Current systems for the analysis of driving behavior are based on data from a variety of sensors, such as vehicle speed, acceleration, braking, and steering. However, due to the rapid development of machine learning, particularly deep learning, it has become feasible to analyze large amounts of time-series data on driving behavior and identify abnormal driving behavior automatically and with a high level of accuracy [15], [18].

It has been shown in recent research works that the data from sensors is significant in detecting the minor patterns of behaviour that are difficult for the drivers to notice. Sensors mounted in the vehicle, smartphone sensors, and the vehicle's on-board diagnostic system have been employed extensively in the study of the dynamics of the vehicle. Deep learning models with the help of recurrent neural networks, convolutional neural networks, or a combination of both have proved effective in detecting the behaviour of the vehicle with high accuracy [15].

Besides model choice, data-related problems like class imbalance have emerged as an important aspect in driving behavior analysis. This is because hazardous driving incidents are less common than normal driving incidents, and thus, imbalanced datasets may affect model performance. To address such limitations, data augmentation methods like Conditional Tabular Generative Adversarial Networks have been proposed to produce realistic data for the less common class to enhance model resilience and generalization capabilities [15]. This chapter offers a broad initiation into the research and covers background, problem statement, research objectives, research methodology, scope, and finally, the organization of the research report, to help readers navigate through the course of research [18].

1.2 PROBLEM STATEMENT

The unsafe driving actions are still not easy to identify correctly due to the fact that there is a lack of accurate systems for analyzing driving actions. Most existing solutions only use simple factors, such as speed and sudden actions, and do not take into consideration the complex actions that take time to develop. This leads to situations where actions like aggressive and drowsy driving are not detected.

In addition, driving behaviour data is time-based, but some existing approaches do not effectively analyse the sequence of driving data over a period of time. This problem is made worse by the imbalance in real-world driving datasets, where normal driving behaviour appears much more often than unsafe behaviour, causing models to produce biased and unreliable results.

1.3 OBJECTIVE

The objectives of this project can be summarized as follows:

1. To identify the limitations and deficiencies in existing driving behaviour analysis systems and understand how data is used to predict unsafe driving behaviour.
2. To design and develop a hybrid deep learning model that combines Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN) techniques for accurate classification of driving behaviours, including normal, aggressive, and drowsy driving.
3. To improve model performance by generating synthetic data for rare driving behaviours using data augmentation techniques, addressing the issue of imbalanced driving datasets.
4. To evaluate the effectiveness of the proposed model by comparing its classification accuracy and risk prediction performance with existing approaches.

1.4 METHODOLOGY

This study follows a structured approach to analyse driving behaviour using deep learning and time series sensor data. The process begins with selecting and preparing driving datasets that include labelled behaviours such as normal, aggressive, and drowsy driving. The raw data is cleaned to reduce noise, scaled to ensure consistency, and divided into fixed length time windows so that changes in driving behaviour over time can be analysed properly.

To analyse driving behaviour that changes over time, a hybrid deep learning model is used that combines Bidirectional Long Short Term Memory and Convolutional Neural Network models. The convolutional layers are used to identify important patterns in short segments of driving data, such as sudden acceleration or braking. The bidirectional recurrent layers then analyse how these patterns develop across a sequence of driving data, allowing the model to understand driving behaviour over a period of time.

Real world driving datasets often contain fewer samples of unsafe driving behaviour compared to normal driving. To reduce this imbalance, data augmentation is applied to increase the number of samples for rare driving behaviours. Synthetic driving data is generated for underrepresented classes before training the model. Finally, the performance of the proposed model is evaluated using standard measures such as accuracy and confusion matrix analysis, and the results are compared with existing methods to determine the effectiveness of the proposed approach.

1.5 SCOPE OF STUDY

This study focuses on analysing driving behaviour using time series sensor data and deep learning techniques, with emphasis on classifying normal, aggressive, and drowsy driving. The work is limited to offline data analysis and model development using labelled driving datasets, without real-time implementation or on-road deployment. This project will use the classification accuracy and confusion matrix analysis to assess the performance of the models. This project will handle the issue of imbalanced classes in the dataset using the concept of data augmentation. Hardware integration, real world data collection, and policy level applications are beyond the scope of this project.

1.6 REPORT ORGANIZATION

This report consists of five chapters. Chapter 1 is the introduction of the report, where an overview, background, problem statement, objectives, methodology, and scope of the study are described. Chapter 2 is the literature study chapter where the related work from the fields of driving behavior analysis, deep learning algorithms, and data augmentation techniques is discussed. Chapter 3 describes the proposed methodology, including data preprocessing, model architecture, and training strategy. Chapter 4 presents the experimental results, performance evaluation, and discussion of findings. Finally, Chapter 5 concludes the study and outlines the limitations and recommendations for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 OVERVIEW

This chapter provides an overview of the current research on the analysis of driving behavior and the identification of unsafe driving practices via data-driven approaches. The chapter will discuss the type of driving data used in the research, the machine learning and deep learning algorithms applied to the research, and the methods of evaluation that have been used in previous studies on the topic. Particular attention will be given to time-series sensor data that has been used to model real-world driving behavior.

Current research studies have utilized machine learning and deep learning techniques in the examination of driver behavior through the use of data from vehicle sensors, OBD systems, and smartphone sensors. The aim is to identify potentially risky driver behaviors, such as aggressive and drowsy driving, through the extraction of data patterns from the data collected from the drivers [18]. However, the variability in the usage of the data and the approaches in the studies has resulted in variations in the accuracy of the existing systems [3], [12].

This chapter aims to describe the limitations and shortcomings of the state-of-the-art driving behavior analysis systems and explain how data is utilized for predicting dangerous driving actions. By reviewing the literature, this survey points out that the important issues, such as the insufficient use of time-variant data, imbalance, and unequal system performance, serve as the basis for the proposed method in this research work [12], [18].

2.2 DRIVING BEHAVIOUR AND UNSAFE DRIVING PATTERNS

Driving behaviour refers to how drivers control their vehicles on the road, regarding gear changes, acceleration, and braking, as well as steering. When you monitor these signals over time, you come out with patterns that describe whether your driving is safe or not. Many studies divide driving behaviour into normal and unsafe categories, with unsafe driving linked to an increased danger of accidents and traffic offenses. This distinction is important because unsafe actions are likely to occur in succession rather than being an isolated incident.

The types of unsafe behaviours require different amounts of data and analysis. For example, driving behavior analysis research may involve analyzing driving patterns obtained from vehicle dynamics and time series sensor data, while other research may combine a variety of information sources to gain a better understanding of the behaviour [12]. For example, driving behavior analysis systems may involve signals for how the vehicle moves and analyzing the sequence of signals in time to identify patterns that indicate unsafe behaviours, as in [12].

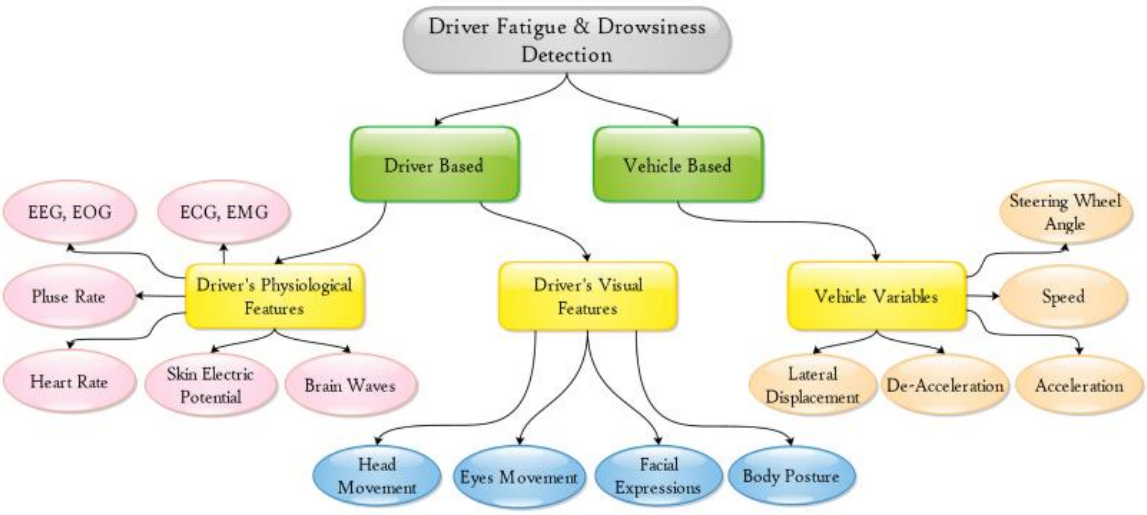

Figure 2.1 Driver-based and vehicle-based features used for fatigue and drowsiness detection [5]

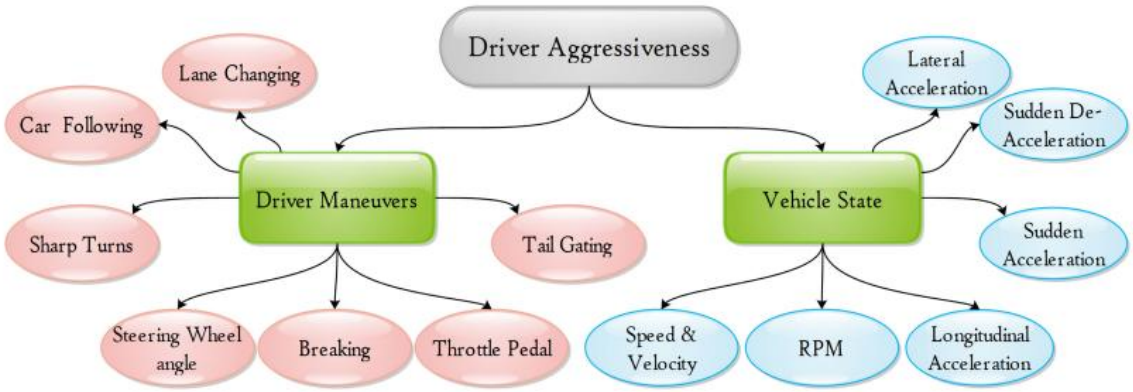

Figure 2.2 Driver maneuvers and vehicle state features used for aggressive driving detection [5]

To strengthen this section visually, the review style diagrams in [5] can also be integrated, such as Figure 2.1 in [5]**,** which summarizes common approaches for detecting driver fatigue, and Figure 2.2 in [5], which highlights the importance and typical methods used for detecting driver aggressiveness. These figures support the explanation that unsafe driving has multiple forms and that each form can be linked to different observable indicators.

Several studies emphasise that unsafe driving behaviour cannot be reliably identified using isolated events alone. Instead, unsafe behaviour must be analysed across a sequence of driving actions, making time series analysis essential. This observation is supported by studies that compare short event-based detection with sequence-based detection and demonstrate improved behaviour recognition when temporal patterns are considered [8], [9]. Essentially, driving behaviour needs to be viewed from a perspective that involves a continuous process, not discrete points.

However, existing studies differ in how unsafe driving behaviours are defined and labelled. Some projects focus on one aspect, for example, stone driving, while others attempt to identify several with the same data set. It all makes for varying levels of data set structure, attribute selection, and performance.

2.3 DRIVING DATA SOURCES AND FEATURE REPRESENTATION

Driving behaviour analysis relies heavily on the quality of driving data and the way features are represented before model training. Existing studies employ a variety of data sources, ranging from real-world naturalistic driving datasets to simulator-based and vision-based datasets. These data sources capture different aspects of driving behaviour, such as vehicle dynamics, driver actions, and environmental conditions, which directly influence the performance of machine learning and deep learning models.

2.3.1 Driving Data Sources

One of the most widely used real-world datasets in driving behaviour research is the UAH-DriveSet, which has been adopted by many studies due to its realistic driving scenarios and comprehensive sensor coverage. The dataset is collected using smartphones installed in vehicles and includes GPS, accelerometer, gyroscope, and speed information recorded during normal, aggressive, and drowsy driving conditions [1], [3], [11]. The naturalistic nature of this dataset allows models to learn realistic driving patterns, making it suitable for real-world deployment.

Aside from using sensor data, most research sources data from image-driven datasets collected through dashboard or in-cabin cameras. The purpose of the datasets is to identify distracted or otherwise errant driving through the analysis of the driver's position, limb movements, and facial expressions [4], [12], [18]. The addition of vision data provides the spatial information missing from data collected through vehicle sensors, particularly for distracted driving.

Some researchers have combined various types of data, such as sensor data and image data, to improve the robustness of systems. Hybrid datasets merge the sensor data with the visual data, allowing for the training of systems capable of learning not only the temporal patterns associated with driving but also the visual patterns associated with the driving behaviors of the drivers [5], [17].

2.3.2 Feature Representation Techniques

Most driving behaviour studies process raw time-series sensor data, such as speed, acceleration, and steering angle, as these signals naturally reflect driving dynamics over time. Sliding window or fixed-length segmentation is commonly applied to transform continuous driving data into sequences suitable for learning temporal patterns [3], [11], [14]. Normalization and noise filtering are frequently used to improve data consistency.

For image-based datasets, feature representation typically involves raw image inputs that are resized and normalized before being passed into convolutional neural networks (CNNs). These models automatically learn spatial features such as hand position, head orientation, and driver posture without manual feature engineering [4], [12].

Several studies also employ statistical or engineered features, such as mean speed, acceleration variance, braking intensity, and turning rate, extracted from raw sensor signals [1], [10]. While these features reduce model complexity, they may fail to capture complex temporal dependencies present in driving behaviour.

The advantages and characteristics of commonly used driving data sources and feature representations from selected studies are summarized in Table 2.1, which is discussed alongside the text for clarity.

Table 2.1 Summary of Driving Data Sources, Feature Representation, and Advantages

Reference	Data Source	Sensor/Data Types	Feature Representation	Advantages
[1]	UAH-DriveSet (real-world)	GPS, accelerometer, gyroscope, speed	Raw time-series with windowing	Realistic driving patterns; suitable for temporal modelling
[3]	UAH-DriveSet (real-world)	Smartphone inertial sensors	Sequential time-series	Captures long-term temporal dependencies
[4]	In-cabin image dataset	Dashboard camera images	Raw images (CNN input)	Effective for detecting visual distraction
[5]	Hybrid dataset	Sensor data + images	Time-series + image features	Multi-modal representation improves accuracy
[11]	Real-world driving data	GPS, speed, vehicle dynamics	Statistical and temporal features	Robust for behaviour classification
[12]	Image-based dataset	Facial and hand imagesRaw time-series with windowing	CNN-based feature learning	Strong performance for distraction detection

2.3.3 Discussion and Research Implications

From the studies reviewed, it is evident that real-world sensor data gives strong timing signals for modeling how people drive, while image-based data shines at spotting distraction-related actions. In many works, however, there is adherence to just one modality, which hurts robustness when conditions change.

Besides, while deep learning takes raw time-series features quite well, many of the studies really do not bring in contextual information-such as road type, weather, traffic flow, or time of day-into the feature mix. This gap demands the need for richer feature representations that combine temporal sensor data with these additional cues.

In general, the literature shows that choosing the appropriate source and representation of data is critical for accurate modeling of driving behavior. The insights support the call for advanced temporal models and better data representations, both of which have been pushed in this project.

2.4 KEY DRIVING BEHAVIOUR PARAMETERS

It is important to understand that the key to grasping the way people drive is to identify the parameters relevant to the way a car is driven and its interactions with the surroundings. In general research and experiments related to driving a car, parameters are classified into two groups. Those related to the basic operations on the vehicle are considered the primary driving parameters. The conditions surrounding the behavior are the other parameters that can be termed as the contextual behavior conditions.

2.4.1 Primary Driving Parameters

Primarily driving parameters recognize the straightforward relationship between a driver and the vehicle they drive. The information is primarily derived from sensors on the vehicle itself or, in some cases, a smartphone the driver might have on them, and primarily includes factors such as the vehicle's velocity, accelerations in the forward or sideways direction, the effort being made on the brakes, the angle of the steering wheel, or the rate of yawing of the vehicle [1], [3], [10].

Studying practical examples, such as the 'UAH-DriveSet' data set, scientists see that the patterns in the time series represented by the data on speed and acceleration can always determine distinct driving patterns, which can be 'aggressive', 'drowsy', or 'normal' driving [1], [3], [11]. 'Aggressive' driving patterns display immediate acceleration in speed, braking, or frequent lane changes, with 'drowsy' drivers preferring relatively steady speeds and slow responses to stimuli [14], [17].

Furthermore, the parameters for controlling the car, such as the angle and turning speed of the steering wheel, also contribute to recognizing behaviors, particularly when combined with analysis related to acceleration values [10], [21]. These parameters provide subtle activities related to controlling the car that could potentially be overlooked when looking at speeding alone. As such, most deep learning models consider the mentioned parameters as raw input, allowing models such as LSTM and BiLSTM to detect long-term dependencies among the activities related to car behavior [3], [12].

Overall, primary driving parameters form the core input features for most machine learning and deep learning models in driving behaviour analysis due to their direct relationship with driver actions and vehicle dynamics.

2.4.2 Contextual Behaviour Conditions

Contextual Behavior: This explains why driving patterns change according to various different circumstances, going beyond what primary driving factors state regarding the control of driving. Variables such as road type, traffic density, pedestrians or animals present in the vicinity, environmental factors like weather and lighting conditions, and adjacent vehicles are taken into consideration to determine how an individual drives, particularly in an urban environment. Many studies emphasize how ignoring context may affect how well a model performs.

With regard to contextual information, most information is of a vision-based type, that is, front cameras or GPS/mapping information. There has been extensive research carried out concerning vision-based detection of traffic items such as cars, pedestrians, road signs, traffic signals, as these items are considered key indicators of driving risk complexity.

The way in which contextual data can be layered and used for driving behavior analysis is illustrated in figure 2.3. Beginning with the perception level and incorporating data generated by vehicle sensors and front cameras, the data is then filtered and prepared at the next level. The final stage involves the detection and classification level where data related to the environment and the driver is analyzed. Several detectors assess the driving scene and danger levels. Finally, the alert level is where everything is combined.

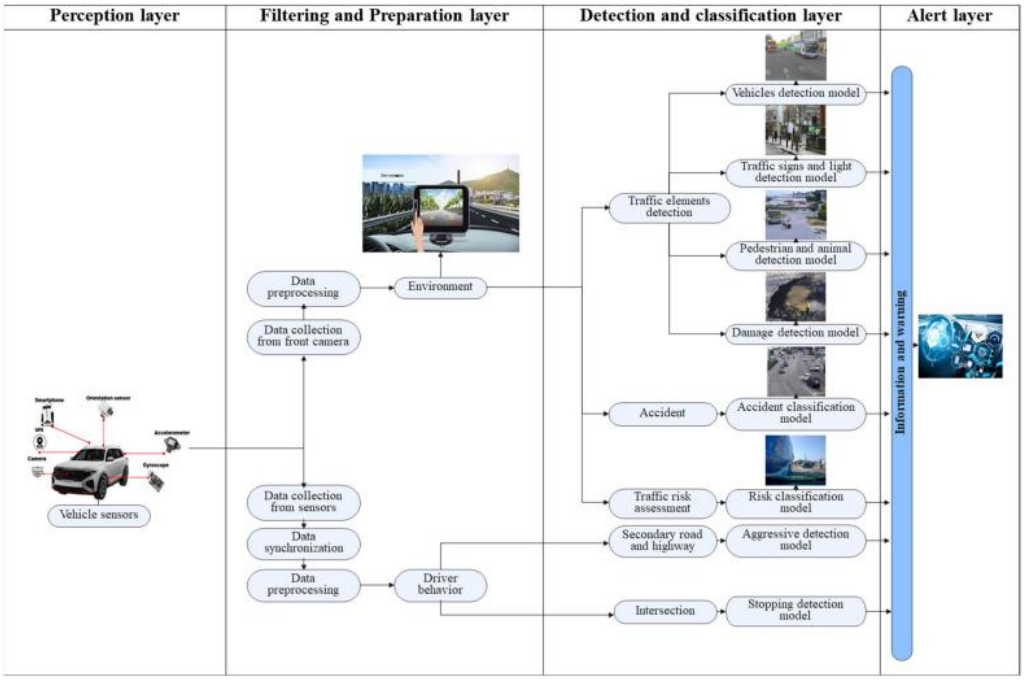

Figure 2.3: Layered Driver and Environmental Context Architecture in SafeSmartDrive [22]

This multi-layered method reflects the trend in current research, where an interplay between the motion of the car and the environmental context is used to more effectively decode the driving pattern [5], [17], [21]. For example, identifying an aggression driving pattern becomes more accurate when you take into consideration secondary roads and highway conditions simultaneously with the patterns of speed and acceleration [11]. Similarly, identifying pedestrians and traffic signals provides vital context for risk estimation analysis in an urban context [4], [18].

Yet, despite these advances, a significant amount of research remains that handles contextual information separately or ignores it altogether due to the increased computation required [2], [22]. This indicates the significance of incorporating contextual behavioral information into basic driving parameters to develop a more realistic system for driving behavior analysis.

2.5 TRADITIONAL MACHINE LEARNING APPROACHES FOR DRIVING BEHAVIOUR ANALYSIS

Traditional machine learning methods have been widely adopted for driving behavior analysis because of their relatively low computational complexity. Before the widespread adoption of deep learning techniques, many research works on driving behavior analysis adopted traditional machine learning classifiers to recognize unsafe driving patterns like aggressive driving, drowsy driving, or abnormally driven patterns using vehicle dynamics data and smartphone sensor data.

The main concentration of the classical ML studies is on the engineered features extracted from the raw driving data, such as the velocity, acceleration, braking torque, and steering variables. The features are usually extracted by utilizing the statistical features of the mean, variance, maximum, and number of events within a pre-defined time window [1], [10], [11]. After the extraction of the features, the classifiers SVM, RF, DT, and kNN are used.

Support Vector Machines are some of the most commonly employed classifiers and are well-known for their strong performance capabilities even when dealing with small and medium-sized datasets and their ability to handle high-dimensional spaces. Theoretical and empirical studies show that SVM-based models can effectively provide accurate results for distinguishing aggressive from normal driving behaviors [10], [21]. The scalability of SVM is limited due to its sensitivity to kernels and parameters.

Random Forest classifiers have been widely used for driving behavior analysis due to their robustness to noise and ability to efficiently process imbalanced datasets. Studies using real-world datasets like UAH-DriveSet show that RF classifiers outperform simple classifiers in efficiently modeling the non-linear associations between the driving variables [1], [11]. In addition, the RF classifier offers the advantage of assessing the relevance of individual driving variables. However, the RF classifier relies heavily on handcrafted features and fails to model time-related dynamics. Methods based on Decision Trees and Ensemble Learning, including Gradient Boosting, have been employed for improving the classification accuracy and maintaining interpretability. These methods are efficient and suitable for real-time processing, though they may pose difficulties in generalizing the patterns of driving behaviors, especially for complex and highly dynamic cases [10], [22]. Likewise, the implementation of the approach based on kNN can be very easy and effective in a controlled setting, though it may become less effective with the growing size of the datasets and the presence of noisy inputs [21].

The key characteristics, advantages, and limitations of commonly used traditional machine learning approaches in driving behaviour analysis are summarised in Table 2.2, which complements the discussion above.

Table 2.2: Traditional Machine Learning Approaches for Driving Behaviour Analysis

References	Machine Learning Model	Data Type Used	Advantages	Disadvantages
[1]	Random Forest	Sensor-based time-series (engineered features)	Robust to noise; handles non-linear patterns; feature importance available	Relies on manual feature extraction; no temporal modelling
[10]	Support Vector Machine (SVM)	Vehicle dynamics and GPS features	Effective on small datasets; strong classification boundary	Sensitive to parameter tuning; limited scalability
[11]	Gradient Boosting	Statistical driving features	High accuracy; handles feature interactions well	Computationally expensive; prone to overfitting
[21]	k-Nearest Neighbour (kNN)	Engineered sensor features	Simple implementation; no training phase	Poor scalability; sensitive to noise
[22]	Decision Tree	Structured driving features	Interpretable; low computation cost	Weak generalization; unstable with complex data

2.5.1 Discussion

From the reviewed literature, it is evident that traditional machine learning approaches provide a solid foundation for driving behaviour analysis, particularly when datasets are limited in size and features are well-defined. These models are suitable for early-stage systems and real-time applications due to their efficiency and interpretability [1], [10].

However, a major limitation across traditional ML methods is their inability to explicitly model temporal dependencies in sequential driving data. Since driving behaviour is inherently time-dependent, relying solely on handcrafted statistical features can result in information loss [3], [12]. This limitation has motivated recent research to shift towards deep learning models that directly learn temporal and spatial representations from raw data.

Nevertheless, traditional machine learning approaches remain an important benchmark and are often used as baseline models for performance comparison in modern driving behaviour analysis systems.

2.6 DEEP LEARNING METHODS FOR TIME SERIES DRIVING BEHAVIOUR ANALYSIS

Deep learning methods have become increasingly dominant in driving behaviour analysis due to their ability to learn complex temporal and spatial patterns directly from raw sensor data. Unlike traditional machine learning approaches that rely on handcrafted statistical features, deep learning models automatically extract discriminative representations from sequential driving signals, making them particularly suitable for time-series driving behaviour analysis [3], [12], [17].

2.6.1 CNN and LSTM Behaviour Recognition

Among deep learning approaches, the most commonly used algorithms for driver behavior recognition are Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. TheCNN is particularly suited to extracting localized features related to the temporal/spatial organization of multivariable sensor data, whereas the LSTM network design focuses on identifying long-term dependencies within a sequence [3], [14].

LSTM networks address the vanishing gradient problem of standard recurrent neural networks by introducing gating mechanisms that regulate information flow. The internal state of an LSTM cell is updated as equations 1, 2, and 3.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C_t} \quad \text{equation(1)}$$

$$\mathrm{o}_t = \sigma(W_{\mathrm{o}}[h_{t-1}, x_t] + b_{\mathrm{o}}), \qquad h_t = \mathrm{o}_t \cdot \tanh(C_t) \quad \text{equation (2),(3)}$$

where f_t, i_t, and o_t denote the forget, input, and output gates, respectively, and C_t represents the cell state. These equations highlight how LSTM models selectively retain or discard historical driving information, which is crucial for modelling behaviour such as aggressive acceleration or prolonged drowsy driving [3], [12].

Several studies further extend LSTM into Bidirectional LSTM (BiLSTM) architectures, enabling the model to learn temporal dependencies in both forward and backward directions. Comparative results reported in prior work demonstrate that BiLSTM consistently outperforms standard LSTM in terms of precision, recall, F1-score, and accuracy across multiple driving behaviour categories [12]. This performance improvement is illustrated in Figure 2.4, where BiLSTM shows more stable and higher classification scores compared to single-direction LSTM, particularly for unhealthy and aggressive driving patterns.

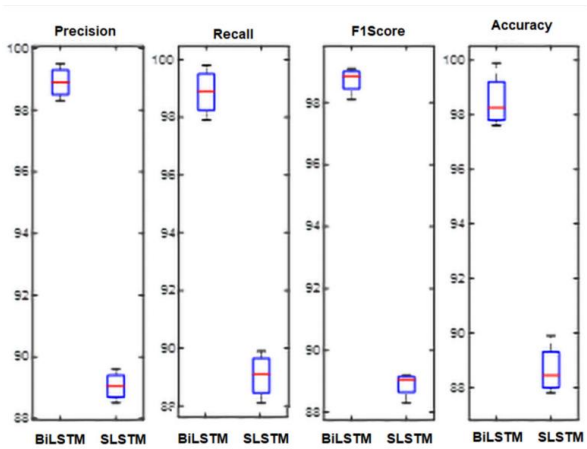

Figure 2.4 Driving behaviour classification BiLSTM vs. SLSTM [12]

CNN–LSTM hybrid models have also been proposed to exploit the strengths of both architectures. In such pipelines, CNN layers are used to extract high-level representations from raw sensor sequences, which are then passed to LSTM layers for temporal modelling [14], [17]. This structure has proven effective for capturing both short-term driving events and long-term behavioural trends.

2.6.2 Hybrid and Generative Approaches

While CNN and LSTM models achieve strong performance, real-world driving datasets often suffer from class imbalance, where unsafe driving behaviours occur far less frequently than normal driving. To address this limitation, recent studies introduce hybrid and generative approaches, particularly Generative Adversarial Networks (GANs), to augment minority driving behaviour classes [12], [17].

Conditional Tabular GAN (CTGAN) has been widely adopted for synthesising realistic time-series or tabular driving data. As illustrated in Figure 2.5, CTGAN consists of a generator G that learns to produce synthetic samples conditioned on class labels and a discriminator D that distinguishes between real and synthetic data. The adversarial training objective can be expressed as equation 4:

$$min_G max_D \; E_{x\sim p_{data}}[logD(x) + \; E_{z\sim p_z}[\log(1 - D\big(G(z|c)\big))] \quad \text{equation (4)}$$

where c represents conditional information such as driving behaviour class. By generating high-quality synthetic samples, CTGAN effectively balances datasets and improves classifier generalisation, especially for aggressive and drowsy driving categories [12].

Hybrid frameworks combining BiLSTM-based classifiers with GAN-based data augmentation have shown significant improvements in classification robustness and stability

[17]. These approaches are particularly suitable for time-series driving behaviour analysis, where collecting large volumes of unsafe driving data is challenging and costly.

The characteristics, advantages, and limitations of representative deep learning approaches are summarised in Table 2.3, which complements the discussion above.

Table 2.3: Deep Learning Approaches for Driving Behaviour Analysis

Reference	**Deep Learning Model**	**Data Type**	**Advantages**	**Disadvantages**
[3]	LSTM	Time-series sensor data	Captures long-term temporal patterns	Single-direction context only
[12]	BiLSTM	Sequential driving data	Higher accuracy; bidirectional temporal learning	Increased computational cost
[14]	CNN–LSTM	Multivariate time-series	Learns spatial and temporal features jointly	Complex architecture
[17]	Hybrid DL (CNN + BiLSTM)	Sensor-based time-series	Strong robustness; improved generalisation	Requires large training data
[12]	CTGAN + DL classifier	Imbalanced tabular data	Handles class imbalance effectively	GAN training instability

The experiments demonstrated that the introduction of synthetic samples enhanced model consistency, especially for difficult-to-detect behaviors. Generally, the hybrid system combining deep learning, streaming decision trees, and generative data augmentation was found to be highly performant, flexible, and suitable for real-time driver monitoring systems.

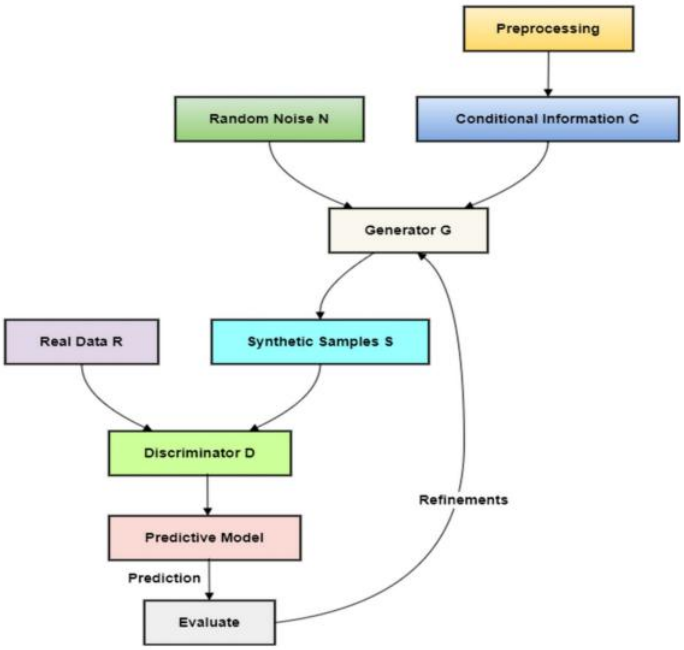

Figure 2.5 Data augmentation process CTGAN [12]

2.6.3 Discussion

Based on the studies that have been reviewed from the literature, the superiority of deep learning over traditional machine learning for time-series driver behavior is clearly evident. This is because such approaches have the ability to learn temporal dependencies from data. Even more advanced models such as those based on BiLSTMs and CNN-LSTMs refine the definition of features.

Moreover, generative approaches such as CTGAN overcome a significant drawback in the analysis of driving behavior, specifically the issue of class imbalance. This is achieved through the generation of lifelike instances in the minority class. Cumulatively, these findings provide significant support for the use of a combination of deep learning and generative models in reliable driving behavior analysis.

2.7 TRANSFER LEARNING AND OBJECT DETECTION

In driver behavior systems, road environment knowledge is as important as driver behavior analysis. Transfer learning is widely used for object detection to quickly fine-tune strong pre-trained models to accurately detect vehicles, traffic signs, and road conditions with minimal data.

Transfer learning has played a critical role in advancing behavior analysis for tasks of object detection like detection of nearby cars, traffic signs, and roadside hazards, even further. Transfer learning enables pre-trained deep models, normally trained on large data like MS COCO or ImageNet, to be fine-tuned to perform driving tasks on small supplementary training data. It reduces training time dramatically without compromising high detection performance, especially for real-time applications.

One of the most vivid instances of this solution is the SafeSmartDrive system presented in [22]. The system employs YOLOv9, a high-end object detection model initially pre-trained on MS COCO and further fine-tuned to recognize seven traffic-related classes: cars, buses, trucks, motorcycles, bicycles, traffic lights, and stop signs. YOLOv9 surpassed its previous versions (YOLOv7 and YOLOv8) on both detection accuracy and processing speed.

Its performance in all these models is encapsulated in the following evaluation comparison:

Table 2.4 Model Evaluation Metrics for YOLOv7, YOLOv8, and YOLOv9 [22]

Models	Precision	Recall	mAP@.5	mAP@[5:.95]
[11]	0.62		0.603	
YOLOv7	0.804	0.681	0.734	0.528
YOLOv8	0.771	0.639	0.718	0.518
YOLOv9	0.831	0.735	0.821	0.627

The confusion matrices for each model offer a detailed view of classification accuracy across different object types. Figure 2.6 displays the matrix for YOLOv9, while Figure 2.7 highlights performance limitations in YOLOv7 and YOLOv8.

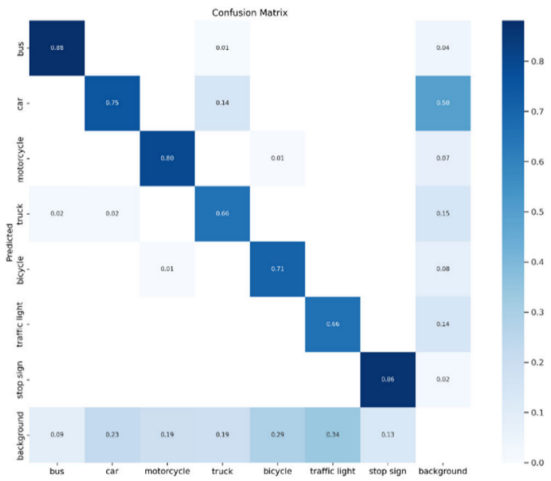

Figure 2.6: YOLOv9 Confusion Matrix [22]

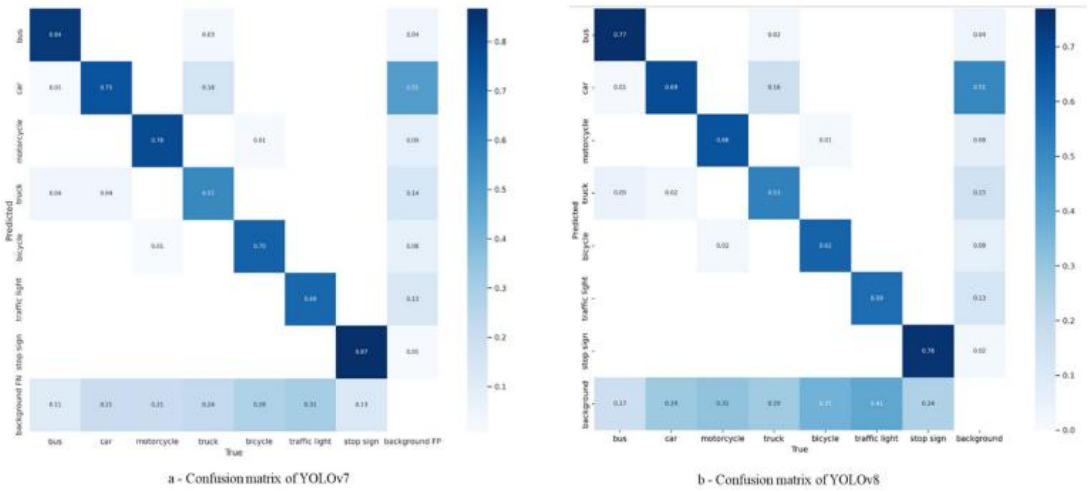

Figure 2.7: YOLOv7 and YOLOv8 Confusion Matrices [22]

Transfer learning for YOLO-based models enables low-overhead training and real-time object detection. It has been demonstrated in [22] that it is possible to get as low as 14 ms detection time for potholes, traffic lights, and stop signs using YOLOv9, which makes it practical for onboard driver assistance and urban hazard detection.

Furthermore, the SafeSmartDrive system possesses a multi-layered architecture that enables it to collect data, prepare the data, detect, classify, and alert in real time. The layering is illustrated in the following system diagram:

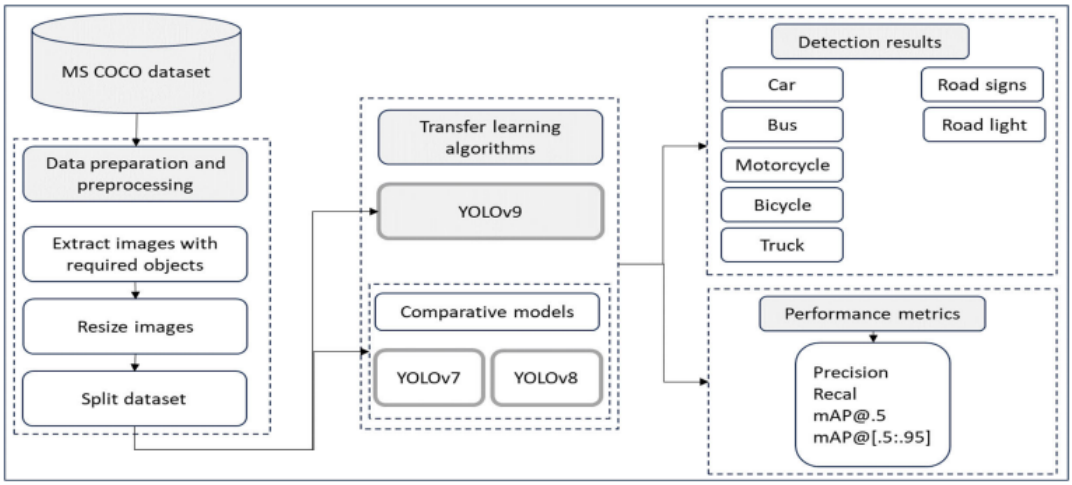

Figure 2.8: SafeSmartDrive System Architecture [22]

The ability of the architecture for feature extraction in both global and local aspects, supported by Dual-Path Network (DPN) modules and spatial pyramid pooling, makes it highly versatile for complicated urban environments. These features are especially critical for inferring traffic context, hazard detection, and providing rich environmental information required in behavior-aware driving analysis systems.

2.8 PREVIOUS WORK ON DRIVING BEHAVIOUR ANALYSIS

Analyzing driver behavior has been a focus area for numerous studies involving a wide range of machine learning and deep learning techniques. Generally, the idea is to identify potentially dangerous behaviors, such as aggressive, drowsy, distracted, or unusual, by analyzing the motion of the car, the actions performed by the driver, and the environment. Various studies adopt different approaches based on the type of input data, feature representations, and models.

In those days, traditional machine learning approaches based on expert-designed features extracted from vehicle sensor data were primarily relied upon by investigators. Corpora like UAH-DriveSet were preferred by many because it resembles ground truth and has good exposure to various vehicle sensors. Classifiers like Random Forest or Support Vectors were quite effective in distinguishing normal and aggressive driving patterns, but feature designs were a barrier to model complexities of temporal patterns.

As a result of increasing availability of sequential driving data, the stage was set for deep learning methods such as architectures based on LSTMs and CNNs. LSTMs were found to be much effective than traditional ML techniques when it comes to longer-term dependencies in driving behaviors. Later on, the concept of BiLSTMs appeared even more promising with their concept of utilizing both past and future information available in a driving pattern for multiple categories of driving.

Recently, there have been attempts to explore hybrid designs of deep learning models such as CNN-LSTM or CNN-BiLSTM to concurrently extract spatial and temporal features from the multi-variate sensor inputs. These hybrid designs are more universally robust with improved generalizability capabilities compared to the single designs. In addition, generative approaches for handling class imbalance, such as the CTGAN method, have been employed to greatly raise the classification accuracy for less common categories of driving behaviors.

To provide a structured comparison of representative studies, Table 2.5 summarises benchmark results from selected prior works, highlighting the models used, datasets, key features, and reported performance metrics.

Table 2.5: Benchmark Table For Driving Behaviour Analysis Using ML

	Study Reference	Model	Dataset	Key Features	Performance
1.	[1]	Random Forest, SVM	UAH-DriveSet	Speed, acceleration, braking, steering (engineered features)	Accuracy ≈ 95%
2.	[3]	LSTM	UAH-DriveSet	Raw time-series sensor data	F1-score ≈ 0.98
3.	[10]	SVM	Real-world driving data	Statistical vehicle dynamics features	Accuracy ≈ 93%
4.	[12]	BiLSTM	UAH-DriveSet	Sequential multivariate sensor data	Up to 96% F1-score
5.	[14]	CNN–LSTM	Smartphone sensor data	CNN-extracted temporal features	Accuracy ≈ 91.9%
6.	[17]	Hybrid CNN–BiLSTM	Real-world driving data	Spatial–temporal driving features	Accuracy > 97%
7.	[12]	CTGAN + DL Classifier	UAH-DriveSet (augmented)	Synthetic + real sensor data	Improved F1-score for minority classes

From the benchmark comparison, it is evident that deep learning models consistently outperform traditional machine learning approaches, particularly for time-series driving behaviour analysis. LSTM and BiLSTM architectures achieve superior performance by learning temporal dependencies directly from raw sensor data, while hybrid CNN-based models further enhance feature representation.

However, several limitations remain in previous works. Many studies either ignore contextual driving conditions or treat them separately, reducing robustness in complex real-world scenarios [2], [21]. Additionally, dataset imbalance remains a common challenge, with only recent works exploring generative augmentation techniques such as CTGAN [12], [17]. These limitations motivate the need for an integrated framework that combines advanced temporal modelling with data augmentation to achieve reliable and generalisable driving behaviour analysis

2.9 SUMMARY

This chapter covers a broad overview of research on driving behavior analysis and is based on where the data originates, feature representation, the key driving features, and the model used. It covers a selection of the various driving datasets, ranging from real-world datasets and sensor data to image-centric datasets and their origins and the significance of such datasets.

It focuses on key driving parameters such as speed, acceleration, braking, and steering angle as key indicators of driving, besides adding that the context in which the drive is happening, such as road type and traffic, also impacts the decisions. While several studies have individually addressed such parameters, there has been constrained research that has simultaneously tackled them.

Though traditional machine learning algorithms such as Support Vector Machines, Random Forests, and Decision Trees perform satisfactorily when trained on manually extracted features, these algorithms fail to provide adequate solutions for the time series patterns found in driving patterns. In contrast to this, the ability of BiLSTM networks to detect long-range patterns from raw data makes them extremely effective for this particular application.

Recent studies reveal that hybrid deep learning models, which integrate CNNs with other LSTM structures, are more effective at modeling driving behavior. These models recognize both spatial and temporal data concurrently, offering an improved feature extraction capability. Another innovative use of deep learning techniques, the CTGAN model, has been exhibited to deal with class imbalance issues by producing authentic samples for the minority class, which improves the robustness of classifiers.

However, a closer analysis of benchmark analysis in existing studies reveals a more complicated scenario. Even in scenarios where accuracy scores and F1 scores seem high, existing studies often struggle with difficulties such as biased data, a lack of context, or difficulties in successfully extending to practical settings. This also underscores a need for a more robust, comprehensive approach where sophisticated modeling of time concepts meets effective data augmentation.

In conclusion, the results above provide a good basis in the current study to support the approach presented in the following chapter. The proposed approach will overcome the weaknesses above by combining a hybrid deep learning model with successful data augmentation to provide precise and reliable results in analyzing the driving behavior.

CHAPTER 3

METHODOLOGY

3.1 OVERVIEW

This chapter explains the methodology used in conducting the research for the analysis of the driving behaviour and how the research was conducted systematically right from the point of investigation to the point of performance analysis. This methodology outlines the entire research process, from design to data analysis.

To achieve the research objectives, this study adopts a structured research methodology that integrates time series analysis of driving data and machine learning methods. The BiLSTM and CNN model is adopted for classifying driving behaviors. Moreover, a data augmentation strategy based on generative approaches is employed to address the class imbalance problem and improve the reliability of the classification results.

This chapter also outlines the tools, software frameworks, and libraries employed during the process of developing the system, in addition to the methods of evaluation that are employed for measuring the performance of the system. On the whole, the proposed methodology ensures that the objectives of the research are linked with the employed techniques, taking into account the limitations that are discussed in the literature, especially the time-dependent behavior modeling and the imbalance of the driving data.

3.2 PROCESS FLOWCHART

The process of designing the driving behaviour analysis system follows a structured project workflow, as shown in Figure 3.1. The flowchart above highlights the activities undertaken throughout the research, from initial exploration to the final system comparison, which is relevant to the research aims.

The proposed project initiates with gathering relevant literature on driving behavior analysis, machine learning, deep learning models, as well as data augmentation. This is done to understand what is already available, what are the deficiencies in current systems, and which benchmark papers to refer to.

With the help of the findings obtained from the literature study, a hybrid deep learning model integrating Bidirectional Long Short-Term Memory (BiLSTM) and Convolutional Neural Network (CNN) architectures is proposed. To start with, the first version of the proposed model is trained using the original driving behavior data without any augmentation. This provides an initial baseline performance for the proposed model.

After the completion of the training process, the performance of the hybrid model is tested using the typical classification performance metrics. This stage yields the results of the performance of the model before the process of data augmentation.

To counter the class imbalance problem in the data, data augmentation is then used to produce a balanced set of data. Synthetic samples for underrepresented classes in driving behavior are produced using generative models. The augmented set of data is then used to retrain the hybrid BiLSTM-CNN model without changing architectures, so that any difference in performance is solely due to data augmentation.

The performance of the hybrid model on the augmented dataset is assessed and compared with the baseline performance achieved prior to performing data augmentation. The effect of data augmentation on classification accuracy and class performance is thereby emphasized.

Finally, the results acquired from the augmented dataset are compared to those in existing benchmark studies and related works. This step is carried out to determine the effectiveness of the proposed approach compared to other models and methods in the literature. The methodology, results, and findings are then written down in the final report. Such a structured project workflow will ensure a thorough assessment of the proposed hybrid model, as well as a clear assessment of the role of data augmentation towards improving driving behavior classification performance

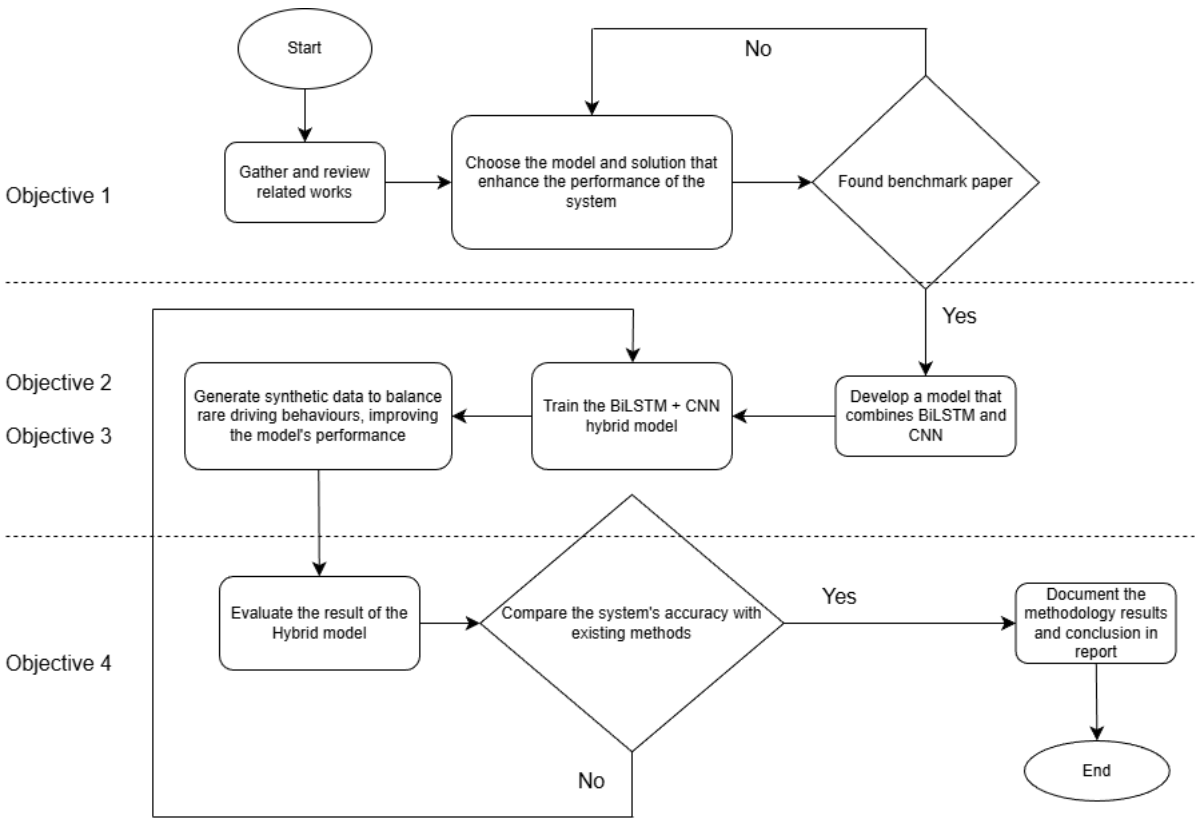

Figure 3.1 Project Flowchart Methodology

3.3 DATASET

This study utilises the UAH-DriveSet, a publicly available dataset developed to support research in driving behaviour analysis. The dataset was collected using a smartphone-based driving monitoring application known as DriveSafe, which records multiple sensor signals during real-world driving sessions [23]. The dataset is designed to facilitate progress in driving behaviour analysis by providing comprehensive time-series data captured under naturalistic driving conditions.

The UAH-DriveSet consists of driving data collected from six drivers with different demographic characteristics and driving styles, using different vehicle models and fuel types. Each driver performed three driving behaviours, namely normal, aggressive, and drowsy driving, across different road environments. The diversity in driver profiles and vehicle types helps improve the generality of the dataset and allows driving behaviour patterns to be analysed under varying conditions.

Table 3.1 summarises the demographic information of the drivers involved in the data collection process, including gender, age range, vehicle model, and fuel type. The dataset includes drivers of different age groups and vehicles powered by diesel, gasoline, and electric fuel sources, which contributes to variation in driving dynamics and sensor measurements.

Table 3.1 List of driver and vehicles that performed the tests [23]

Driver	Gender	Age Range	Vehicle Model	Fuel type
D1	Male	40-50	Audi Q5 (2014)	Diesel
D2	Male	20-30	Mercedes B180 (2013)	Diesel
D3	Male	20-30	Citroen C4 (2015)	Diesel
D4	Female	30-40	Kia Picanto (2004)	Gasoline
D5	Male	30-40	Opel Astra (2007)	Gasoline
D6	Male	40-50	Citroen C-Zero (2011)	Electric

In total, the dataset contains 40 driving trips, corresponding to more than 500 minutes of recorded driving data. Each trip includes raw sensor measurements and additional semantic information. Although video recordings of the trips are provided as part of the dataset, this study focuses exclusively on sensor-based data for driving behaviour classification.

3.3.1 Data Source and Description

The driving data were collected along two predefined routes. The first route is a motorway, with an approximate round-trip distance of 25 km, typically consisting of three lanes in each direction and a maximum permitted speed of 120 km/h. The second route is a secondary road, with a total distance of approximately 16 km, generally consisting of one lane in each direction and a maximum permitted speed of 90 km/h. These two road types allow the dataset to capture driving behaviour under different traffic and speed conditions.

Two types of raw data files from the dataset are used in this study. The first file contains raw accelerometer and orientation data, including acceleration along the x, y, and z axes, as well as roll, pitch, and yaw measurements. The second file contains raw Global Positioning System (GPS) data, including vehicle speed, latitude, longitude, altitude, vertical accuracy, horizontal accuracy, course, and course change. These sensor signals provide detailed information on vehicle motion and driving dynamics, making them suitable for time-series driving behaviour analysis.

The dataset includes three driving behaviour classes: normal driving, drowsy driving, and aggressive driving. The distribution of samples across these classes is shown in Table 3.2. The dataset is imbalanced, with normal driving having the highest number of samples, followed by drowsy and aggressive driving. This imbalance reflects real-world driving conditions, where unsafe behaviours occur less frequently, and motivates the application of data augmentation techniques in later stages of this study.

Table 3.2 Distribution of Driving Behaviour Classes in the UAH-DriveSet [23]

Driving Behaviour	Number of Datasets
Normal	131 866
Drowsy	99 852
Aggressive	79 667

3.3.2 Data Preprocessing and Cleaning

The raw UAH-DriveSet data are provided in plain text file format and do not include explicit driving behaviour labels within the sensor data files. Therefore, the first preprocessing step involves assigning behaviour labels to the data based on the trip-level annotations provided by the dataset. Each data segment is labelled as normal, aggressive, or drowsy driving according to the corresponding driving session.

After labelling, the raw text files are converted into comma-separated values (CSV) format. This conversion simplifies data handling and improves compatibility with data processing libraries, making the dataset easier to clean, process, and analyse.

Data preprocessing and cleaning are then performed to ensure data consistency and quality. Incomplete records and noisy samples are removed, and all numerical features are normalised to reduce scale differences between sensor measurements. Normalisation helps stabilise model training and prevents features with larger numeric ranges from dominating the learning process.

Several features that are not directly related to driving behaviour classification are removed during preprocessing. These include latitude, longitude, altitude, vertical accuracy, and horizontal accuracy, which primarily describe geographic positioning rather than driving dynamics. Removing these features reduces input dimensionality and allows the model to focus on motion-related sensor signals that better represent driving behaviour.

Since the accelerometer and GPS sensors operate at different sampling rates, interpolation is applied to synchronise both data sources. This ensures that all sensor signals are temporally aligned and can be analysed together within the same time window.

After preprocessing and synchronisation, the dataset is split into training and validation sets using a 70:30 ratio. This split allows the model to be trained on a larger portion of the data while maintaining an independent validation set for performance evaluation. The overall class distribution is preserved to ensure a fair assessment of model performance.

To illustrate the structure of the processed data used in this study, Figure 3.2 presents a sample of the dataset after preprocessing. The table shows selected sensor features along with the corresponding driving behaviour label. This sample demonstrates that the data are cleaned, normalised, and properly labelled before being used for window segmentation and model training.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	acc_x	acc_y	acc_z	acc_x_kf	acc_y_kf	acc_z_kf	roll	pitch	yaw	speed_inte	difcourse_	behaviour	trip_id
2	0.674419	-0.27778	0.625	0.25	0.275862	0.8	0.75	0.452381	0.028146	-1.43012	-0.56794	normal	D1_T1
3	1.348837	0.222222	0.65625	1	-0.06897	0.8	0.772727	0.380952	0.028146	-1.43012	-0.56794	normal	D1_T1
4	1.488372	-0.41667	0.90625	1.75	-0.17241	0.8	0.818182	0.428571	0.027594	-1.43012	-0.56794	normal	D1_T1
5	0.627907	-0.41667	0.875	1.785714	-0.31034	1.04	0.75	0.428571	0.027594	-1.43012	-0.56794	normal	D1_T1
6	-0.04651	-0.44444	1.3125	0.857143	-0.55172	1.36	0.704545	0.380952	0.027594	-1.43012	-0.56794	normal	D1_T1
7	0.116279	-0.25	0.53125	0.357143	-0.2069	1.2	0.704545	0.380952	0.028146	-1.43012	-0.56794	normal	D1_T1
8	0.44186	-0.69444	0.90625	0.428571	-0.82759	0.96	0.659091	0.547619	0.028698	-1.43012	-0.56794	normal	D1_T1
9	0.139535	-0.55556	0.5625	0.357143	-0.7931	0.96	0.636364	0.52381	0.028698	-1.43012	-0.56794	normal	D1_T1
10	0.069767	-1.72222	1.0625	0.214286	-1.27586	0.92	0.636364	0.47619	0.029249	-1.43012	-0.56794	normal	D1_T1
11	0.046512	-1.27778	0.78125	0.142857	-1.51724	1.16	0.659091	0.47619	0.030353	-1.43012	-0.56794	normal	D1_T1
12	0.627907	-1.16667	0.6875	0.464286	-1.62069	1.04	0.681818	0.5	0.030905	-1.43399	-0.56794	normal	D1_T1
13	-0.37209	-0.66667	0.4375	0.214286	-1.44828	0.8	0.636364	0.52381	0.031457	-1.43748	-0.56794	normal	D1_T1
14	-0.09302	-0.77778	0.3125	-0.10714	-1.06897	0.52	0.613636	0.642857	0.032009	-1.44135	-0.56794	normal	D1_T1
15	0.604651	-0.13889	0.5625	0.107143	-0.41379	0.72	0.568182	0.619048	0.032561	-1.44522	-0.56794	normal	D1_T1

Figure 3.2: Sample of preprocessed Driving Behaviour Dataset

3.3.3 Window Segmentation Strategy

Driving behaviour is a continuous process that evolves over time rather than a single isolated event. To capture the temporal characteristics of driving actions, the preprocessed sensor data are segmented into fixed-length time windows before being used for model training and evaluation.

In this study, each driving trip is divided into overlapping windows using a sliding window approach. Each window consists of 45 consecutive time steps (SEQ_LEN = 45), which represent a short segment of driving behaviour. The window is shifted forward by 5 time steps (STEP = 5) to create overlapping segments. This overlap increases the number of training samples and ensures that important driving patterns are not missed between consecutive windows.

Window segmentation allows the model to analyse short sequences of driving actions, such as acceleration, braking, and steering changes, while preserving the temporal order of sensor signals. The use of overlapping windows also helps the model learn gradual transitions in driving behaviour rather than abrupt changes.

For each window, a driving behaviour label is assigned based on the dominant behaviour exhibited during that time interval. Each window is labelled as normal, aggressive, or drowsy according to the corresponding driving session. This window-level labelling enables the model to learn behaviour patterns at a fine temporal resolution and supports detailed performance evaluation.

The same segmented windows are used to construct both input representations of the proposed model. The sequential input preserves the raw time-series sensor signals within each window, while the statistical input summarises the same window using descriptive features. This ensures consistency between inputs and allows the model to capture both short-term motion patterns and overall driving characteristics.

Overall, the window segmentation strategy plays a critical role in enabling effective time-series learning and forms the foundation for the hybrid deep learning architecture described in the following sections.

3.3.4 Feature Construction

In order to get the correct driving pattern, a two-input feature construction method is employed by the research. This technique breaks down every segmented window into two different inputs: sequential features and statistics features. By incorporating two inputs, the model can recognize both the minute time-dependent pattern movement and the overall driving pattern of every segmented window.

3.3.4.1 *Sequential Features (X_seq)*

The sequential input, denoted as Xseq, consists of raw time-series sensor signals extracted directly from each segmented window. This input preserves the original temporal order of driving actions, such as acceleration changes, steering movements, and speed variations.

For each window, the sequential input has a shape of:

$$X_{seq} \in \mathrm{R}^{SEQ_LEN \times n_features} \quad \text{equation (5)}$$

where SEQ_LEN = 45 represents the number of time steps in a window, and n_features corresponds to the number of selected sensor signals, including acceleration components, orientation angles, and vehicle speed.

This representation allows the model to learn short-term and long-term temporal dependencies, such as sudden braking, continuous acceleration, or gradual loss of vehicle control, which are important indicators of aggressive and drowsy driving behaviour.

3.3.4.2 *Statistical Features (X_stat)*

In addition to raw time-series data, statistical features, denoted as Xstat, are extracted from the same window to summarise overall driving behaviour. These features describe the distribution and variability of sensor signals within a window and provide complementary information that may not be fully captured by sequential learning alone.

For each sensor signal, the following statistical measures are computed:

- **Mean**, representing the average driving intensity
- **Standard deviation**, indicating variability in driving actions
- **Minimum and maximum values**, capturing extreme behaviour
- **Jerk**, representing the rate of change of acceleration

The mean value of a feature within a window is calculated as:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i \quad \text{equation (6)}$$

where x_i is the sensor value at time step i, and N is the total number of time steps in the window.

The standard deviation, which reflects driving stability, is computed as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2} \quad \text{equation (7)}$$

Jerk is derived as the first-order difference of acceleration and is used to capture sudden changes in vehicle motion, which are commonly associated with aggressive driving behaviour.

The statistical input for each window is represented as:

$$X_{stat} \in R^{(n_{stats})} \quad \text{equation (8)}$$

where n_stats is the total number of extracted statistical features.

3.3.4.3 *Importance of the Two-Input Design*

The combination of sequential and statistical features forms the core strength of the proposed model. The sequential features enable the model to learn fine-grained temporal patterns, while the statistical features provide a compact summary of overall driving style within each window. By integrating both representations, the model gains a more complete understanding of driving behaviour, improving robustness and classification accuracy.

This feature construction strategy ensures that all relevant driving information is effectively represented before being processed by the hybrid deep learning architecture described in the next section.

3.4 PROPOSED MODEL ARCHITECTURE

This section describes the architecture of the proposed driving behaviour analysis system and explains how the different components of the model work together to perform accurate behaviour classification. The model is designed to process time-series driving data and extract meaningful patterns that represent different driving behaviours. By combining multiple learning mechanisms, the proposed architecture can capture both detailed temporal patterns and overall behavioural characteristics within driving data.

The proposed system employs a hybrid deep learning architecture that integrates convolutional, recurrent, and fully connected layers. This architecture is specifically designed to address the challenges identified in previous studies, including the time-dependent nature of driving behaviour and the difficulty of representing complex driving patterns using a single model type. The following subsection provides a high-level overview of the hybrid architecture and explains the motivation behind its design.

3.4.1 Overview of the Hybrid Architecture

The proposed driving behaviour classification model adopts a hybrid deep learning architecture that processes driving data through two complementary input branches and combines their learned representations using late feature fusion. This design is selected to capture both detailed time-dependent driving events and overall behavioural characteristics within each driving window.

As illustrated in Figure 3.3, the first branch is a sequence branch that receives the windowed raw time-series input X_{seq} with shape (45, $n_features$). The input is first normalised to stabilise learning, followed by SpatialDropout1D to improve robustness against noisy sensor channels. A Conv1D layer is then applied to learn local temporal patterns such as rapid acceleration changes or sudden braking. These features are subsequently processed by stacked BiLSTM layers to model longer temporal dependencies within the same window, producing a compact representation of driving behaviour from sequential signals.

In parallel, the second branch is a statistical feature branch that processes X_{stat}, which contains descriptive statistics computed from the same window. These statistics include mean, standard deviation, minimum and maximum values, and jerk-related measures. The branch uses normalisation and fully connected layers to learn behavioural indicators that summarise overall driving style and stability.

After both branches extract meaningful features, their outputs are combined using a concatenation fusion layer, as shown in Figure 3.3. The fused representation is passed through a classification head to produce class probabilities for normal, aggressive, and drowsy driving. Focal loss is used during training to reduce the impact of class imbalance and improve learning for minority behaviour classes.

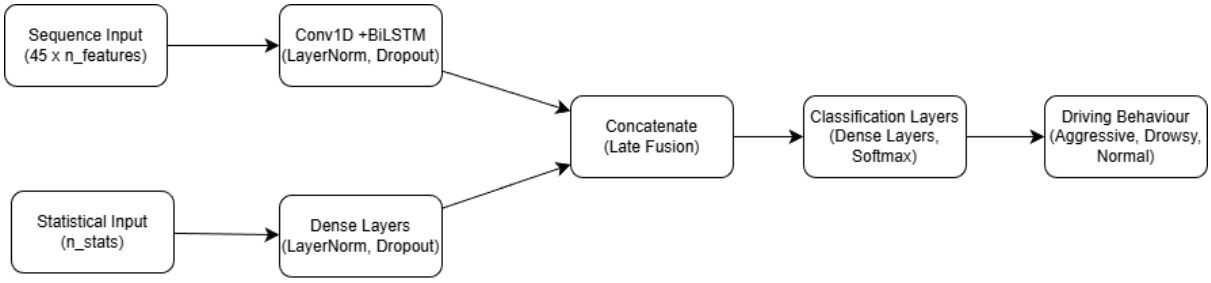

Figure 3.3 Block diagram of the proposed hybrid Conv1D–BiLSTM and statistical feature fusion architecture

3.4.2 Sequence Branch (Conv1D + BiLSTM)

The sequence branch is designed to learn time-dependent driving patterns directly from raw sensor signals arranged as time-series data. This branch processes the sequential input X_{seq}, which preserves the temporal order of driving actions within each window, allowing the model to capture how driving behaviour evolves over time.

The Conv1D layer is applied as the first learning component in the sequence branch. Its main purpose is to extract local temporal features from the input signals. By sliding convolutional filters across the time dimension, the Conv1D layer detects short-term driving events such as sudden acceleration, sharp braking, or rapid steering changes. These local patterns are important indicators of unsafe driving behaviour and act as low-level features that simplify the learning task for subsequent layers.

Following convolution, the extracted features are passed to Bidirectional Long Short-Term Memory (BiLSTM) layers. The BiLSTM is responsible for modelling temporal dependencies within the driving window. Unlike standard LSTM models that process data in a single direction, BiLSTM analyses the sequence in both forward and backward directions. This enables the model to consider what happened before and after a particular driving action, which is essential for understanding behaviours that develop gradually, such as drowsiness or sustained aggressive driving.

Through this combination, the sequence branch performs hierarchical temporal learning. The Conv1D layer focuses on short-term motion patterns, while the BiLSTM layers capture longer-term relationships and contextual dependencies across the entire window. Together, they produce a compact and informative representation of driving behaviour that reflects both instantaneous events and overall temporal trends, making the sequence branch a key component of the proposed hybrid architecture.

3.4.3 Statistical Feature Branch

The statistical feature branch is designed to capture global driving behaviour characteristics within each time window by processing hand-crafted statistical features. While the sequence branch focuses on detailed temporal patterns, this branch provides a complementary view by summarising the overall driving style and variability in a compact form.

The input to this branch is the statistical feature vector X_{stat}, which contains descriptive measures such as mean, standard deviation, minimum and maximum values, and jerk-related statistics computed from the same windowed sensor data. These features describe how stable or aggressive the driving behaviour is over the window duration and are particularly useful for distinguishing consistent patterns such as smooth driving versus highly variable or erratic behaviour.

The statistical features are processed through fully connected (dense) layers with nonlinear activation functions. The dense layers learn meaningful combinations of the input statistics and transform them into higher-level behavioural indicators. This allows the model to capture relationships between multiple statistical measures, such as how large speed variation combined with frequent high jerk values may indicate aggressive driving.

By summarising the driving behaviour at the window level, the statistical feature branch improves the robustness of the overall model. It reduces sensitivity to short-term noise in sensor signals and provides stable behavioural cues that support classification when temporal patterns alone are insufficient. The output of this branch is a compact representation of overall driving style, which is later fused with the temporal features learned by the sequence branch to form a comprehensive behaviour prediction.

3.4.4 Feature Fusion and Classification Head

After feature extraction in both the sequence branch and the statistical feature branch, the learned representations are combined using a late feature fusion strategy. Late fusion is performed by concatenating the high-level features extracted independently by each branch. This approach allows each branch to learn specialised information before integration, resulting in a more stable and expressive feature representation.

The fused feature vector contains both temporal information from the Conv1D–BiLSTM sequence branch and behavioural summaries from the statistical feature branch. This combined representation is passed through a fully connected classification head, which learns high-level relationships between the fused features and driving behaviour classes. Dropout is applied in this stage to reduce overfitting and improve generalisation.

The final layer of the classification head uses a Softmax activation function to produce class probabilities for the three driving behaviour categories: normal, aggressive, and drowsy. The Softmax function converts the raw model outputs into a probability distribution, ensuring that the sum of probabilities across all classes equals one. The Softmax function is defined as:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_i}} \quad \text{equation (9)}$$

Where z_i is the output score for class i, and C is the total number of behaviour classes. The predicted class corresponds to the highest probability value. This probabilistic output enables confidence-based analysis and supports later evaluation using metrics such as precision, recall, and confusion matrices.

3.4.5 Loss Function

Driving behaviour datasets are often imbalanced, where unsafe behaviours such as aggressive and drowsy driving appear less frequently than normal driving. In such cases, standard loss functions may cause the model to favour the majority class, leading to poor detection of rare but critical behaviours.

To address this issue, Focal Loss is used as the training loss function. Focal Loss modifies the standard cross-entropy loss by reducing the contribution of easy-to-classify samples and placing more emphasis on hard and misclassified samples. This encourages the model to focus on minority classes and difficult driving patterns. The focal loss function is expressed as:

$$FL(p_t) = -a(1 - p_t)^{\gamma} \log(p_t) \quad \text{equation (10)}$$

Where p_t is the predicted probality for the true class, a is a weighting factor, and γ is the focusing parameter. In this study, the focusing parameter is set to γ = 2.0, which increases the penalty for misclassified minority samples.

By using focal loss, the proposed model improves learning for underrepresented driving behaviours and reduces classification bias toward normal driving. This contributes to more balanced performance across all behaviour classes and enhances the reliability of the driving behaviour analysis system.

3.5 DATA AUGMENTATION STRATEGY

Datasets describing driving behaviors may have class imbalance when it comes to driving behaviors, where normal driving dominates the minority class of risky driving behaviors, leading to biased models toward the majority class, ignoring extreme but very essential behaviors such as aggressive or sleepy driving.

The augmentation process using the CTGAN model is illustrated in Figure 3.4. The preprocessed real driving data along with the class labels is given as input to the generator along with random noise. The generator produces the synthetic driving data exhibiting the statistical properties of the real driving data. Later on, the discriminator makes an attempt to discriminate between the real and the synthesised data. As a result of continuous improvement efforts, the generator tends to generate high-quality driving data that matches the real driving patterns [12].

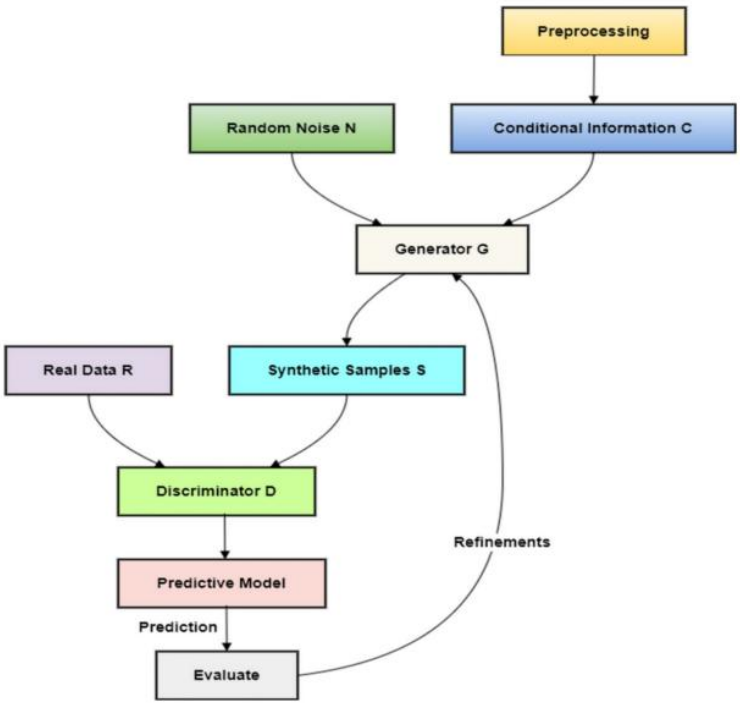

Figure 3.4: CTGAN-based data augmentation framework for generating synthetic driving behaviour samples [12]

The synthetic samples generated by CTGAN are used only during the training phase. First, a baseline model is trained using the original dataset to establish reference performance. Next, synthetic samples are added to the training set to improve class balance, while the validation dataset remains unchanged. This ensures that performance improvements are attributed to better learning rather than data leakage.

To illustrate the quality and structure of the generated synthetic data, Figure3.5 and 3.6 present a sample of synthetic driving data of aggressive and drowsy behaviour produced by CTGAN. The figure shows that the synthetic samples maintain realistic feature values and follow the same structure as the original dataset, indicating that CTGAN successfully captures the underlying data distribution.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	acc_x	acc_y	acc_z	acc_x_kf	acc_y_kf	acc_z_kf	roll	pitch	yaw	speed_inte	difcourse_	trip_id	type_of_r	type_of_r	label	behaviour
2	1.272727	0.820513	0.162791	0.941176	0.243243	0.4	0.039216	0.767442	-0.82591	-0.27995	0.907418	D2_T6	0	1	0	aggressive
3	0.431818	-0.30769	-0.27907	0.147059	-0.24324	-0.48571	-0.35294	0.232558	0.45127	0.847826	1.206323	D1_T6	0	1	0	aggressive
4	-0.38636	-0.89744	-0.65116	-0.08824	-1.05405	0.114286	0.921569	-3.02326	0.106432	-0.26139	0.983651	D5_T6	0	1	0	aggressive
5	-1.15909	0.128205	0.72093	-0.55882	0	1.171429	0.313725	0.418605	2.024023	-0.57017	0.238838	D4_T6	1	0	0	aggressive
6	-2.97727	0.641026	-0.34884	-1.67647	0.837838	-1.05714	0.509804	0.976744	-1.1823	-0.07879	1.511357	D5_T6	0	1	0	aggressive
7	-0.15909	-1.05128	-0.5814	-0.05882	-0.13514	-0.34286	-0.03922	-0.25581	-1.17804	-0.37307	-0.18972	D1_T4	1	0	0	aggressive
8	-0.65909	-0.28205	-0.81395	-0.55882	-0.97297	-1.2	0.235294	-2.04651	0.032842	0.177518	-0.48878	D4_T3	0	1	0	aggressive
9	-0.59091	-0.79487	-0.2093	-0.29412	0.135135	0.4	-0.62745	0.116279	0.029801	0.103591	0.519428	D3_T5	0	1	0	aggressive
10	-0.90909	0.948718	0.093023	-0.82353	0.72973	-3.70771	0.196078	-7.4186	-0.12224	0.196282	-0.35759	D6_T5	1	0	0	aggressive
11	0.636364	0.615385	0.674419	-0.23529	-0.32432	0.142857	0.176471	1.186047	-0.70549	-1.52176	0.476461	D4_T3	0	1	0	aggressive
12	-1.59091	-5.33333	-2.11628	-1.76471	-0.45946	-0.97143	0.411765	0.116279	-0.18732	0.597465	-0.00277	D3_T7	1	0	0	aggressive
13	0.113636	-0.15385	-0.37209	0.382353	1.108108	0.2	0.372549	0.069767	2.135299	1.388131	-0.04956	D5_T2	1	0	0	aggressive

Figure 3.5: Sample Synthetic Aggressive Behaviour Driving Data Generated Using CTGAN

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	acc_x	acc_y	acc_z	acc_x_kf	acc_y_kf	acc_z_kf	roll	pitch	yaw	speed_inte	difcourse_	trip_id	type_of_r	type_of_r	label	behaviour
2	0.473684	-0.31707	0.424242	0.137931	-0.51351	0.047619	0.236842	-0.71053	0.567222	0.252944	-0.49621	D6_T4	0	1	1	drowsy
3	-0.86842	0.365854	-0.75758	-0.03448	0.432432	0.142857	-0.10526	0.447368	1.236876	0.862878	-0.02465	D2_T3	1	0	1	drowsy
4	0.026316	1.04878	-0.12121	-1.96552	0.675676	-0.28571	0.157895	-0.44737	-0.28041	-0.20044	0.723457	D3_T6	0	1	1	drowsy
5	-0.18421	0.731707	-5.33333	-0.37931	0.135135	-5	-3.63158	-0.36842	-1.04289	0.216385	-0.02794	D3_T2	1	0	1	drowsy
6	1.684211	-0.85366	-0.87879	2.241379	-0.02703	-0.19048	1.473684	0.526316	0.847631	0.074952	-0.32773	D4_T4	0	1	1	drowsy
7	-1.05263	-0.43902	0.969697	-0.27586	-1.24324	-0.42857	0.736842	-0.97368	-2.06338	1.021008	0.093368	D1_T5	1	0	1	drowsy
8	-0.44737	2.146341	-0.66667	-3.37931	-2	-1.61905	0.368421	5.921053	1.65749	-2.67543	10	D5_T3	1	0	1	drowsy
9	-1.18421	-0.2439	-1.18182	-0.65517	-0.72973	-1.95238	0.447368	0.131579	-0.35595	0.254723	-0.17072	D4_T7	1	0	1	drowsy
10	0.131579	-0.65854	-0.18182	0.310345	-0.35135	0.190476	-0.44737	-1.5	0.615237	0.281858	0.167274	D1_T7	0	1	1	drowsy
11	0.263158	0.780488	-1	-0.41379	0.351351	-0.38095	-0.44737	-0.68421	-0.20038	0.707942	-0.31646	D1_T7	0	1	1	drowsy
12	-0.86842	0.731707	0.30303	1.241379	0.594595	0.52381	-0.05263	-0.21053	-1.98656	0.35735	-0.70356	D1_T5	1	0	1	drowsy
13	0.763158	-0.2439	0.181818	1.068966	-2.45946	-0.57143	0.131579	0.026316	-0.06082	-0.50892	1.949968	D4_T4	0	1	1	drowsy
14	2.552632	1.073171	0.848485	0.758621	0.378378	1.952381	-1.26316	0.868421	-0.04673	-5.65756	-1.50029	D4_T4	0	1	1	drowsy
15	0.447368	1.146341	-0.60606	0.034483	0.162162	-0.47619	0.315789	-0.92105	0.475032	-0.51833	1.736076	D4_T4	0	1	1	drowsy

Figure 3.6: Sample Synthetic Drowsy Behaviour Driving Data Generated Using CTGAN

3.6 MODEL TRAINING AND EVALUATION SETUP

This section describes the training configuration and evaluation procedure used to assess the performance of the proposed driving behaviour analysis model. The setup is designed to ensure fair evaluation, prevent data leakage, and provide reliable performance estimates at the window level.

3.6.1 Training Configuration

The model is trained using a Stratified Group K-Fold cross-validation strategy. This approach is selected to preserve the class distribution across folds while ensuring that data from the same driving trip does not appear in both training and validation sets. Grouping by trip prevents information leakage caused by overlapping windows extracted from the same driving session. In this study, 5-fold cross-validation is applied, where the dataset is divided into five mutually exclusive folds, and each fold is used once as validation data while the remaining four folds are used for training.

The training process is configured for a maximum of 30 epochs per fold. To reduce overfitting and unnecessary computation, early stopping is employed by monitoring the validation loss. Training is automatically stopped when the validation performance no longer improves for a predefined number of epochs, and the model weights corresponding to the best validation performance are restored. This strategy ensures stable convergence while preventing the model from memorising training data.

3.6.2 Window-Level Evaluation

Model evaluation is performed at the window level, where each segmented time window is treated as an independent sample. This evaluation strategy directly reflects the model's ability to classify short-term driving behaviour patterns based on sensor data. Window-level evaluation is suitable for analysing the effectiveness of temporal feature learning and provides fine-grained insight into classification performance across different behaviour classes.

Predictions from all validation folds are aggregated to compute overall performance metrics. This cross-validated evaluation provides a robust estimate of model generalisation and reduces bias caused by a single train–validation split.

3.6.3 Evaluation Metrics

To comprehensively evaluate classification performance, several standard metrics are used, including accuracy, precision, recall, and F1-score. Accuracy measures the proportion of correctly classified windows and is defined as:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad \text{equation (11)}$$

Where TP, TN FP and FN represent true positives, true negatives, false positives, and false negatives, respectively.

Precision evaluates how many predicted positive samples are correctly classified:

$$Precision = \frac{TP}{TP+FP} \quad \text{equation (12)}$$

Recall measures the ability of the model to correctly identify actual positive samples:

$$Recall = \frac{TP}{TP+FN} \quad \text{equation (13)}$$

The F1-score provides a balanced measure by combining precision and recall:

$$F1\ score = \frac{2\ x\ Precision\ x\ Recall}{Precision + Recall} \quad \text{equation (14)}$$

In addition, confusion matrix analysis is used to visualise class-wise prediction performance and identify common misclassification patterns between normal, aggressive, and drowsy driving behaviours. Together, these metrics provide a comprehensive evaluation of the proposed model's effectiveness under imbalanced and time-dependent driving data conditions.

3.7 SYSTEM FLOWCHART

Figure 3.7 illustrates the overall system flow of the proposed driving behaviour analysis methodology. The process begins with the acquisition of raw accelerometer and GPS data from the UAH-DriveSet. The collected data is first labelled and converted into a structured format before undergoing preprocessing and cleaning to remove unnecessary features and normalise sensor values. The accelerometer and GPS signals are then synchronised and interpolated to ensure consistent temporal alignment, followed by window segmentation to generate fixed-length samples suitable for model input.

Next, feature construction is performed by extracting sequential time-series features and computing statistical features from each window. The hybrid Conv1D–BiLSTM model is initially trained using the original dataset to establish baseline performance, which is evaluated using window-level metrics such as accuracy, precision, recall, and F1-score. To address class imbalance, data augmentation is applied using CTGAN, and the model is retrained using the augmented dataset. Finally, the augmented model is evaluated and its performance is compared with the baseline results, completing the system workflow.

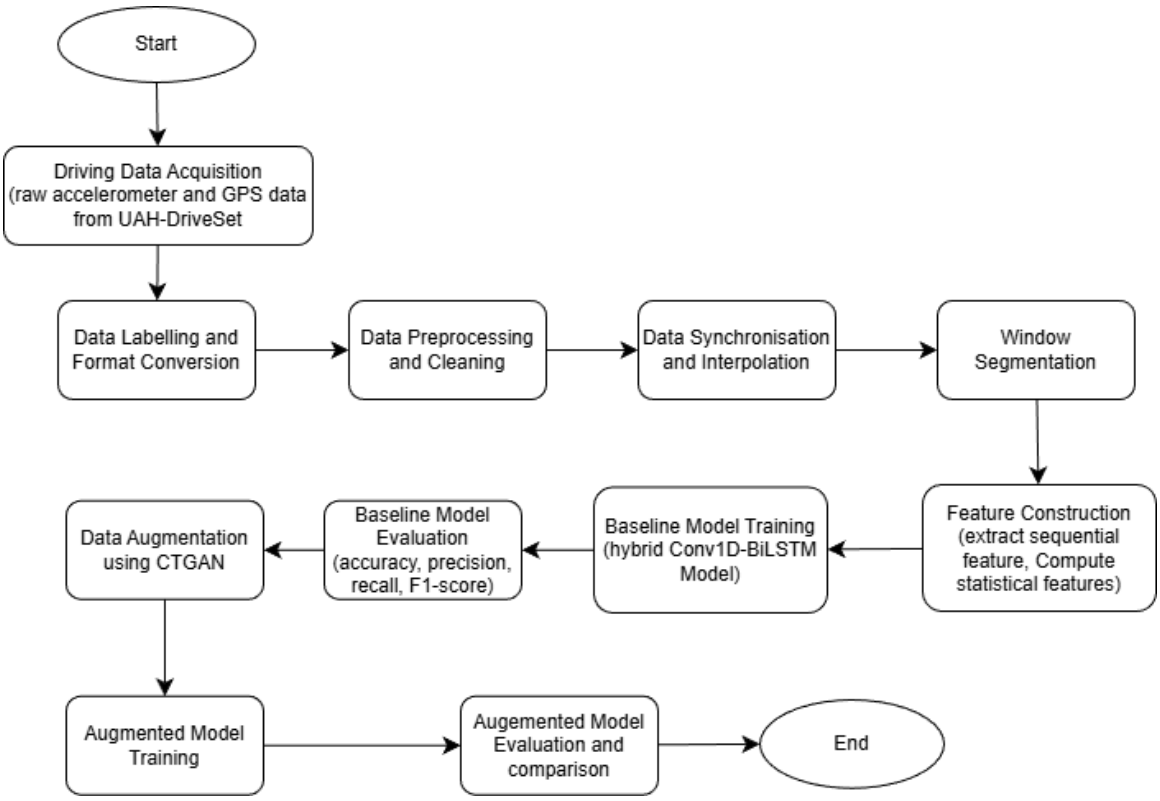

Figure 3.7 Flowchart propose solution

3.8 DEVELOPMENT TOOLS AND ENVIRONMENT

This section describes the development environment and tools employed in developing the driving behavior analysis system. The tools provide ease of development, model training, and testing.

3.8.1 Programming Languages and Libraries

This system is developed mainly in Python, currently probably the most-used language in machine learning and data science because of its versatility, cleanliness of syntax, and strong ecosystem of libraries.

The key libraries in Python help in managing and controlling the entire process involved in the system, from preparing the data to testing its efficiency. The key libraries include Numpy and Pandas. They handle large time-series data and perform all operations such as windowing and feature extraction very efficiently. They put all things in their proper places.

Scikit-learn is employed for data splitting, feature scaling, and model evaluation. In particular, it is used to implement the Stratified Group K-Fold cross-validation strategy, ensuring balanced class distribution while preventing data leakage across training and validation sets. Scikit-learn also provides standard evaluation metrics that are used to assess classification performance.

For model development and training, TensorFlow and Keras are used as the primary deep learning frameworks. These libraries facilitate the implementation of the proposed hybrid Conv1D–BiLSTM architecture, including convolutional layers, recurrent layers, feature fusion mechanisms, and the use of custom loss functions such as focal loss to address class imbalance.

Matplotlib and Seaborn are utilised for data visualisation and result presentation. These tools are used to generate training and validation loss curves, confusion matrices, and comparative performance plots, which aid in analysing model behaviour and interpreting experimental results.

In addition, CTGAN-related libraries are used to generate synthetic driving data for minority behaviour classes. These libraries enable the creation of realistic synthetic samples that improve class balance during training, contributing to more stable and reliable model performance.

These libraries can be combined with a focus towards rapid experimentation, effective reproducibility, and thorough evaluation of the new method being validated. By incorporating reliable, publicly available tools, this system can easily be extended or replicated by others in the future.

3.9 SUMMARY

This chapter discusses the overall strategy that is adopted to develop and test the system design proposed for the analysis of driving behavior. The strategy is commenced with the description of the dataset and the data preparation phases of labeling and pre-processing the data. Subsequent phases of pre-processing the data are discussed next and then the feature development section is progressed into the system design related to two-input design concepts.

The next topic covered in the discussion is the hybrid deep learning model, which explains the combination of Conv1D, BiLSTM, and fully connected layers, as well as the methods used in feature fusion and classification. To overcome the imbalance problem and improve the identification of rare behaviors, the concept of focal loss and data augmentation with CTGAN has been covered in this chapter. Training, validation, and evaluation criteria, along with the use of evaluation metrics, ensure a fair evaluation.

Finally, towards the end, the chapter has been provided with a list of the software development tools and software environment used in this study. The methodology introduced here provides a comprehensive framework to perform behavior analysis to overcome the challenges faced in the current state-of-the-art methods introduced in literature.

The next chapter shall introduce how the methodology introduced above has been used to provide experiment results analyzing the performance of the proposed system.

CHAPTER 4

RESULT AND ANALYSIS

4.1 OVERVIEW

This chapter presents and analyses the experimental outcomes based on our driving behavior analysis system. It is essential that we discuss here that the main aim of this experimentation is to evaluate the effectiveness of the proposed hybrid deep learning model in classifying driving behaviors at the window level. To assess this, we conduct an experiment that would help in understanding its baseline, understanding how data augmentation based on CTGAN affects it, and by comparing it with other models in existing literature.

4.2 EXPERIMENTAL SETUP

This section describes the experimental configuration used to evaluate the proposed driving behaviour classification model. The experimental setup was carefully designed to ensure fair evaluation, reliable performance measurement, and prevention of data leakage caused by the temporal and grouped nature of driving data.

4.2.1 Dataset Partitioning

The dataset consists of multiple driving trips, where each trip contains time-series sensor data labelled with driving behaviour classes. Since the data is collected continuously over time, multiple overlapping windows are extracted from the same trip. If these windows are randomly split into training and validation sets, windows from the same trip may appear in both sets, leading to data leakage and overly optimistic results.

To address this issue, StratifiedGroupKFold cross-validation was employed. This method was chosen because it ensures that all windows belonging to the same trip are assigned exclusively to either the training or validation fold, thereby preventing information leakage across folds. At the same time, stratification was applied to maintain a balanced distribution of driving behaviour classes in each fold. This is particularly important because the dataset is imbalanced, with minority behaviours such as aggressive and drowsy driving appearing less frequently than normal driving.

A total of five folds (N = 5) were used. For each fold, four subsets were used for training, and one subset was used for validation. This configuration provides a good balance between computational efficiency and reliable performance estimation, while allowing the model to be evaluated on multiple unseen driving trips.

4.2.2 Window Segmentation

Each driving trip was segmented into overlapping windows using a sliding window approach. This method was chosen because driving behaviour is not instantaneous, but instead evolves over a short period of time. A fixed window length of 45 samples was selected to capture sufficient temporal information required to characterise driving behaviour within each segment.

A step size of 5 samples was used to generate overlapping windows. This overlap increases the number of training samples and allows gradual transitions in driving behaviour to be captured more smoothly. However, overlapping windows can introduce label ambiguity near behaviour change boundaries. To reduce this issue, a label purity threshold of 0.95 was applied. A window was retained only if at least 95% of its samples belonged to the same driving behaviour class. This decision helps ensure that each window represents a dominant and consistent driving behaviour, leading to more reliable window-level labels and more stable model training.

4.2.3 Input Features

Two types of input representations were used for each window to capture complementary aspects of driving behaviour. The first input consists of sequential features**,** which are raw time-series sensor signals extracted from the driving data. These features preserve the temporal order of the signals and were chosen to allow the model to learn fine-grained driving dynamics such as acceleration changes, speed variations, and short-term motion patterns that are critical for behaviour classification.

The second input consists of statistical features, which are hand-crafted descriptors computed from each window. These features include the mean, standard deviation, minimum, and maximum values of each signal, as well as the mean and standard deviation of the first-order difference (jerk). Statistical features were included because they summarise the overall behaviour within a window and provide global contextual information that may not be easily captured by sequential learning alone. By combining sequential and statistical features, the model can learn both local temporal patterns and overall driving behaviour characteristics.

4.2.4 Model Training Configuration

The proposed hybrid model was trained using a consistent configuration across all experiments to ensure fair comparison It was trained for a period of 30 epochs, which was an appropriate training time that ensured it learned enough without overfitting due to substantial training. The size of the batch was set at 256, which was responsible for ensuring that the gradients were stable.

The choice of the optimizer was made to be AdamW because it combines the fast convergence properties of the Adam optimizer with the decoupled weight decay, which promotes generalization. The learning rate was chosen to be 3×10^{-4} because it is standard for similar complexity models. The focal loss function with $\gamma = 2.0$ was chosen because it helps the network learn more from the hard examples by giving less importance to the easier ones.

To enhance stability and generalization further, several regularizers were utilized in this project, including dropout, L2 regularization, and gradient clipping. Learning rate scheduling and early stopping were employed to avoid overfitting in the network while steering its optimization towards finding a best solution.

4.3 BASELINE MODEL PERFORMANCE (WITHOUT DATA AUGMENTATION)

This section presents the performance of the proposed hybrid CNN–BiLSTM model when trained using the original dataset without data augmentation. The evaluation focuses on window-level classification results obtained across all five cross-validation folds. These results serve as the baseline for assessing the effectiveness of the proposed architecture before introducing CTGAN-based data augmentation.

4.3.1 Window-Level Classification Performance

This section presents the window-level classification performance of the proposed hybrid deep learning model evaluated across all folds. Window-level evaluation measures the model's ability to correctly classify short driving segments and provides insight into how well local driving behaviour patterns are recognised.

Table 4.1 summarises the precision, recall, and F1-score for each driving behaviour class. The overall classification accuracy achieved by the model is 75.6%, indicating that the proposed architecture is capable of learning meaningful patterns from time-series driving data.

For the **Aggressive driving class (Class 0)**, the model achieved a precision of **88.6%**, recall of **69.2%**, and an F1-score of **77.7%**. The high precision suggests that when the model predicts aggressive driving, the prediction is usually correct. However, the comparatively lower recall indicates that a portion of aggressive driving windows are misclassified as other behaviours, suggesting that aggressive behaviour may share overlapping temporal characteristics with other classes.

For the **Drowsy driving class (Class 1)**, the model obtained a precision of **73.1%**, recall of **76.2%**, and an F1-score of **74.7%**. The balanced precision and recall values indicate that the model can recognise drowsy driving behaviour more consistently than aggressive driving. This may be due to more stable and distinguishable temporal patterns present in drowsy driving behaviour.

For the **Normal driving class (Class 2)**, the model achieved a precision of **71.8%**, recall of **79.0%**, and an F1-score of **75.2%**. The higher recall reflects the model's strong ability to correctly identify normal driving windows. However, the slightly lower precision suggests that some windows from aggressive and drowsy driving are misclassified as normal driving, which is expected due to the higher frequency and broader behavioural characteristics of normal driving.

Overall, the model achieved a **macro-average F1-score of 75.9%**, demonstrating relatively balanced performance across all driving behaviour classes despite the presence of class imbalance. These baseline results confirm the effectiveness of the proposed hybrid architecture, while also highlighting the need for further improvement in minority class recognition.

Table 4.1 Baseline Model CNN-BILSTM Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.886	0.692	0.777	15840
1 (Drowsy)	0.731	0.762	0.747	19848
2 (Normal)	0.718	0.790	0.752	26226
Overall Accuracy			0.756	61914
Macro Usage	0.779	0.748	0.759	61914

4.3.2 Confusion Matrix Analysis Baseline Model

Figure 4.1 presents the window-level confusion matrix obtained across all cross-validation folds. The confusion matrix provides detailed insight into the classification behaviour of the model and highlights the patterns of misclassification between different driving behaviours.

For the **Aggressive driving class**, a total of **10,965 windows** were correctly classified. However, **1,059 windows** were misclassified as drowsy driving and **3,816 windows** were misclassified as normal driving. This distribution explains the lower recall observed for aggressive driving, as a significant number of aggressive windows are confused with other behaviours, particularly normal driving.

For the **Drowsy driving class**, the model correctly classified **15,129 windows**, with **402 windows** misclassified as aggressive driving and **4,317 windows** misclassified as normal driving. The relatively small number of misclassifications into aggressive driving indicates that the model can distinguish drowsy behaviour from aggressive behaviour reasonably well. However, confusion with normal driving remains noticeable.

For the **Normal driving class**, **20,723 windows** were correctly classified. Nevertheless, **1,007 windows** were misclassified as aggressive driving, and **4,496 windows** were misclassified as drowsy driving. Although the model shows strong recall for normal driving, it also absorbs misclassified windows from other classes, which contributes to its lower precision.

Overall, the confusion matrix reveals that most misclassifications occur between behaviour classes that share similar short-term temporal characteristics. This outcome is expected in window-level driving behaviour analysis and highlights the challenge of separating behaviours with overlapping motion patterns. These observations further justify the application of data augmentation and advanced feature learning techniques to improve class discrimination in subsequent experiments.

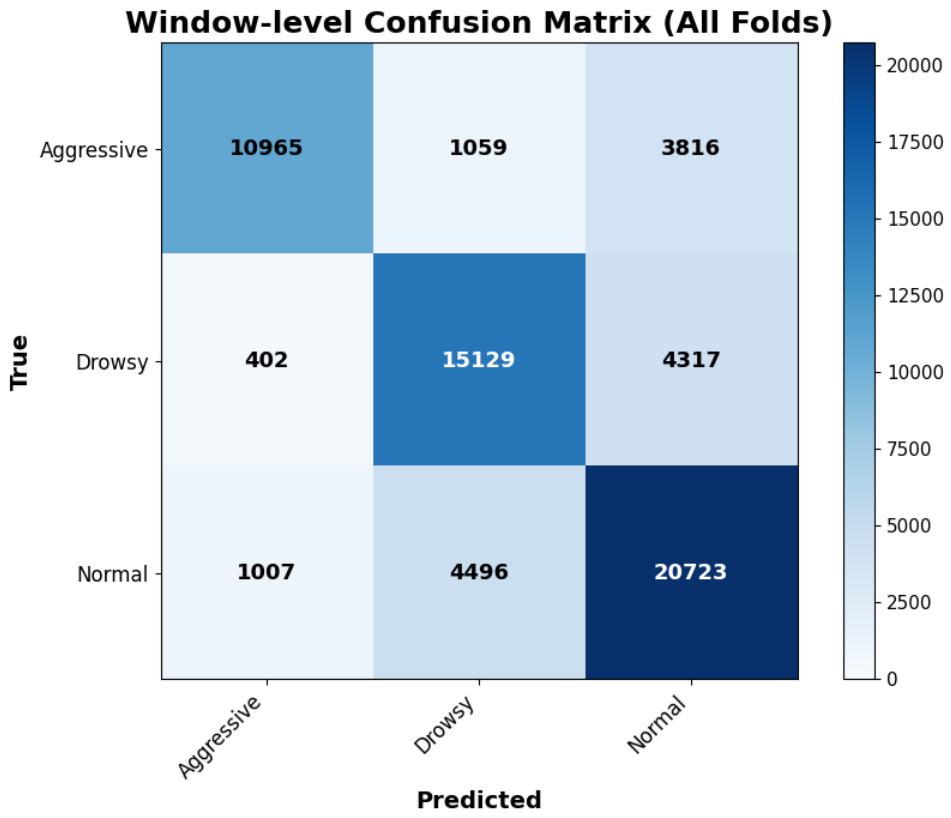

Figure 4.1: Confusion Matrix Analysis Baseline Model

4.4 IMPACT OF DATA AUGMENTATION ON MODEL PERFORMANCE

This section evaluates the effect of CTGAN-based data augmentation on the window-level classification performance of the proposed driving behaviour analysis system. The main objective of this evaluation is to determine whether augmenting minority driving behaviour samples can improve model generalisation and reduce the impact of class imbalance observed in the baseline results.

4.4.1 Performance After Data Augmentation

After applying CTGAN-based data augmentation, the proposed hybrid CNN–BiLSTM model demonstrated a clear improvement in window-level classification performance. The overall classification accuracy increased to **80.4%**, compared to **75.6%** achieved by the baseline model without data augmentation. This improvement indicates that the augmented training data helped the model learn more representative and discriminative patterns across different driving behaviours.

For the **Aggressive driving class**, the model achieved a precision of **93.7%**, recall of **82.7%**, and an F1-score of **87.9%**. Compared to the baseline results, both recall and F1-score show a substantial increase, indicating that the model is now able to identify aggressive driving windows more effectively. The improvement in recall suggests that fewer aggressive driving windows are being misclassified as other behaviours, which is a direct benefit of increasing the number of aggressive samples during training.

For the **Drowsy driving class**, the model obtained a precision of **78.4%**, recall of **81.3%**, and an F1-score of **79.8%**. These values represent a consistent improvement over the baseline performance, particularly in recall. This indicates that the model is better able to recognise drowsy driving behaviour after augmentation, likely because CTGAN-generated samples increased the diversity of drowsy behaviour patterns seen during training.

For the **Normal driving class**, the model achieved a precision of **71.2%**, recall of **77.2%**, and an F1-score of **74.1%**. While the improvements for the normal class are smaller compared to the minority classes, the performance remains stable. This is expected, as the normal driving class was already well represented in the original dataset. Importantly, the performance gain for aggressive and drowsy driving was achieved without significantly degrading the performance of the normal class.

Overall, the model achieved a **macro-average F1-score of 80.6%**, reflecting balanced performance across all driving behaviour classes. These results demonstrate that CTGAN-based data augmentation effectively improves model robustness and generalisation, particularly for minority driving behaviours.

Table 4.2 Augmented Baseline Model Classification Report

Class	**Precision**	**Recall**	**F1-score**	**Support**
0 (Aggressive)	0.937	0.827	0.879	27253
1 (Drowsy)	0.784	0.813	0.798	27037
2 (Normal)	0.712	0.772	0.741	26226
Overall Accuracy			0.804	80516
Macro Usage	0.811	0.804	0.806	80516

4.4.2 Confusion Matrix Analysis After Data Augmentation

Figure 4.2 presents the window-level confusion matrix obtained after applying CTGAN-based data augmentation. The confusion matrix provides detailed insight into how the model's classification behaviour changes after augmentation and helps identify improvements in class separation.

For **Aggressive driving**, **22,543 windows** were correctly classified, while **1,179 windows** were misclassified as drowsy driving and **3,531 windows** were misclassified as normal driving. Compared to the baseline results, the number of aggressive windows misclassified as other behaviours is noticeably reduced. This improvement explains the significant increase in recall for aggressive driving after data augmentation.

For **Drowsy driving**, the model correctly classified **21,976 windows**, with **410 windows** misclassified as aggressive driving and **4,651 windows** misclassified as normal driving. The reduced misclassification into aggressive driving indicates better separation between aggressive and drowsy behaviours. However, some confusion with normal driving remains, which is expected due to overlapping temporal characteristics between drowsy and normal driving.

For **Normal driving**, **20,235 windows** were correctly classified, while **1,112 windows** were misclassified as aggressive driving and **4,879 windows** were misclassified as drowsy driving. Although normal driving still absorbs some misclassified windows from other classes, the overall pattern shows improved balance compared to the baseline confusion matrix.

Overall, the confusion matrix after data augmentation shows a clearer diagonal structure, indicating improved classification confidence and reduced misclassification for minority classes. These results confirm that CTGAN-based data augmentation improves the model's ability to distinguish between different driving behaviours at the window level.

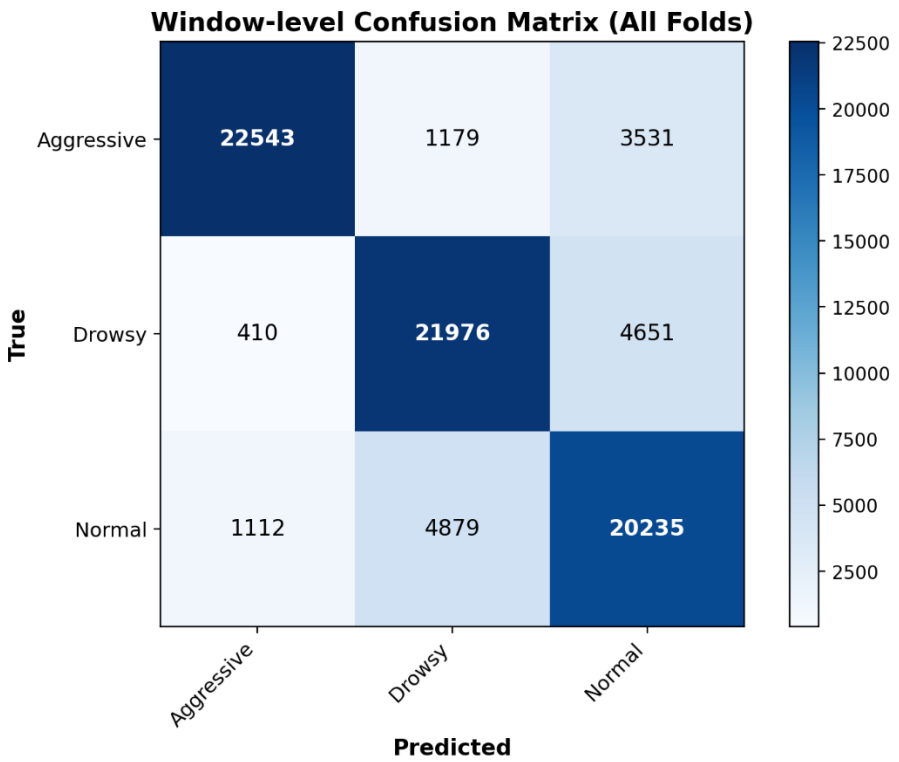

Figure 4.2: 1.1.1 Confusion Matrix Analysis After Data Augmentation

4.4.3 Training Behaviour Across Folds

To further analyse the effect of CTGAN-based data augmentation on model learning, the training and validation accuracy curves for all five cross-validation folds were examined. Figures corresponding to Fold 1 through Fold 5 illustrate the evolution of model accuracy over training epochs for both training and validation sets.

Across all folds, the training accuracy shows a rapid increase during the early epochs, followed by a gradual and stable improvement as training progresses. This behaviour indicates that the proposed hybrid CNN–BiLSTM model is able to quickly learn meaningful representations from the augmented dataset. The steady rise in training accuracy suggests effective optimisation and sufficient model capacity to capture driving behaviour patterns.

The validation accuracy curves closely follow the training accuracy trends in all folds, with validation accuracy consistently remaining slightly higher or comparable to training accuracy. This observation is important, as it indicates that the model does not overfit the training data despite the increased dataset size after augmentation. The small gap between training and validation accuracy suggests good generalisation performance on unseen validation windows.

In Fold 1 and Fold 2, the validation accuracy steadily increases and stabilises around the later epochs, reaching values above 80%. Similar trends are observed in Folds 3 and 4, where validation accuracy improves consistently with minimal fluctuation. Fold 5 shows slightly lower final accuracy compared to other folds; however, the training and validation curves remain smooth and stable, indicating consistent learning behaviour without significant instability.

Across all folds, no sharp divergence between training and validation curves is observed. This confirms that the use of regularisation techniques such as dropout, focal loss, early stopping, and learning rate reduction effectively controls overfitting. Furthermore, the consistent convergence behaviour across folds demonstrates that CTGAN-based data augmentation contributes to more stable training dynamics by providing a more balanced and informative dataset.

Overall, the training curves indicate that the proposed model converges reliably across different data splits. The consistent learning patterns observed in all five folds provide strong evidence that the augmented model is robust, stable, and capable of generalising well to unseen driving data.

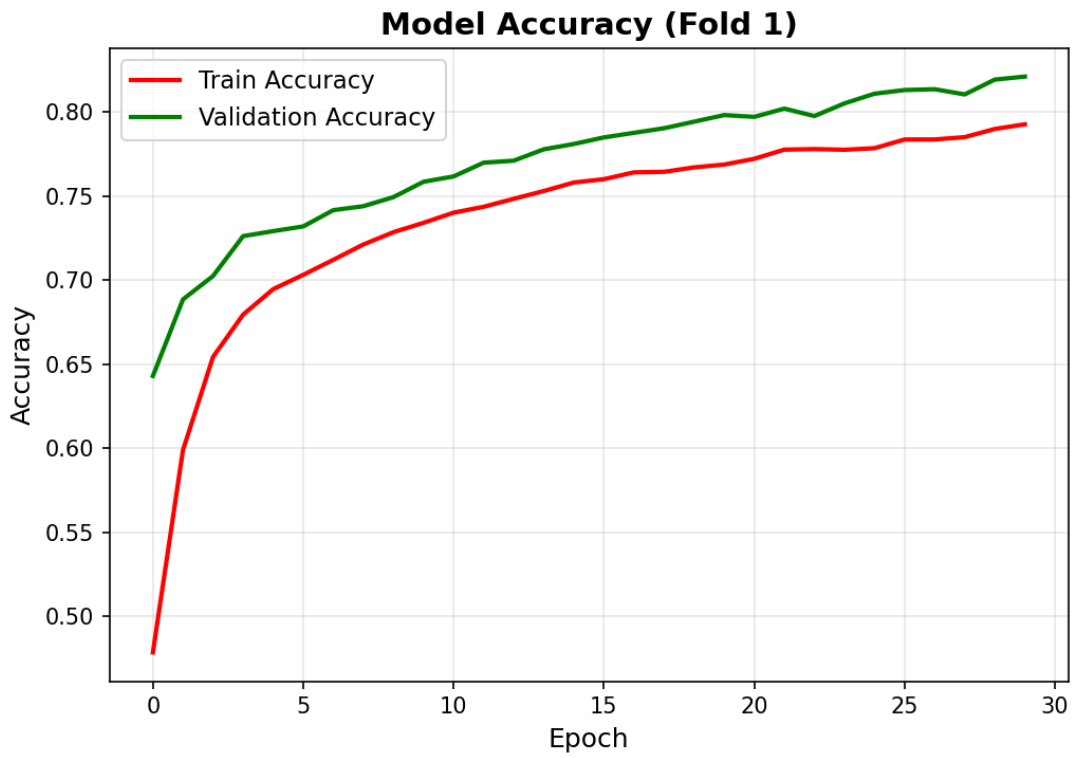

Figure 4.3: Model Accuracy Fold 1

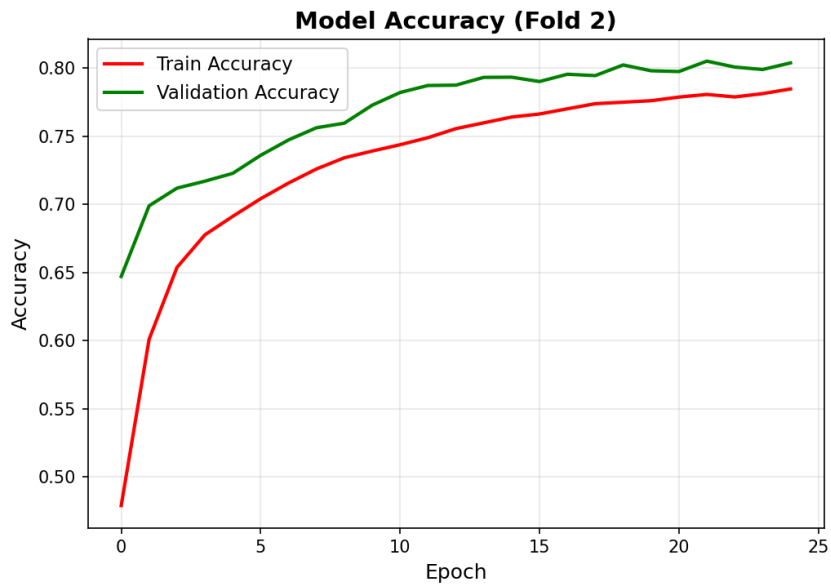

Figure 4.4: Model Accuracy Fold 2

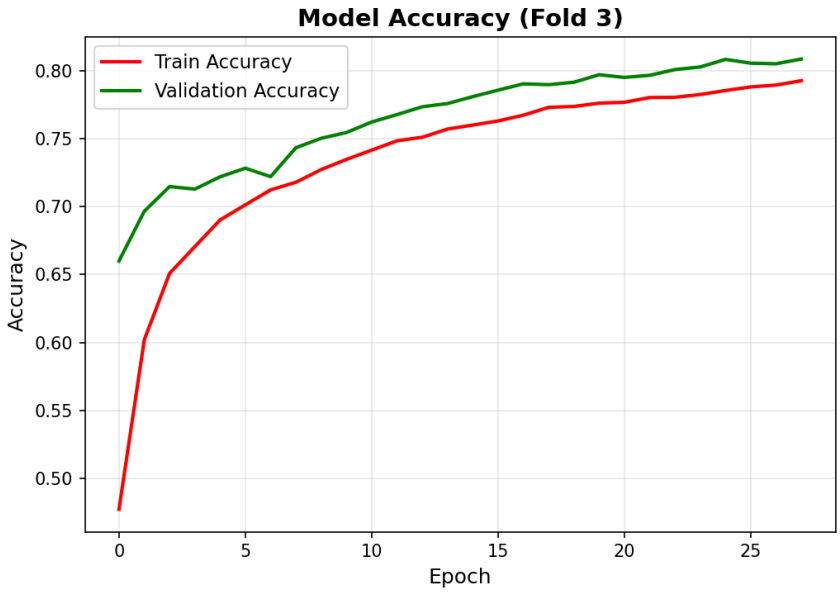

Figure 4.5: Model Accuracy Fold 3

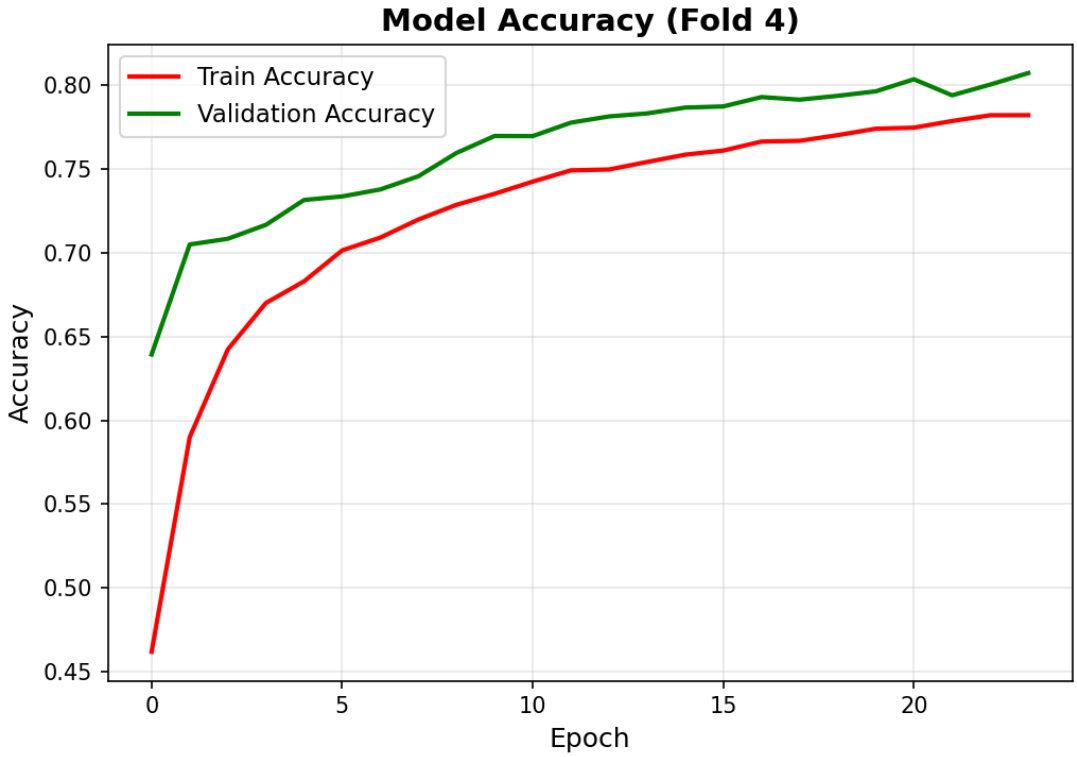

Figure 4.6: Model Accuracy Fold 4

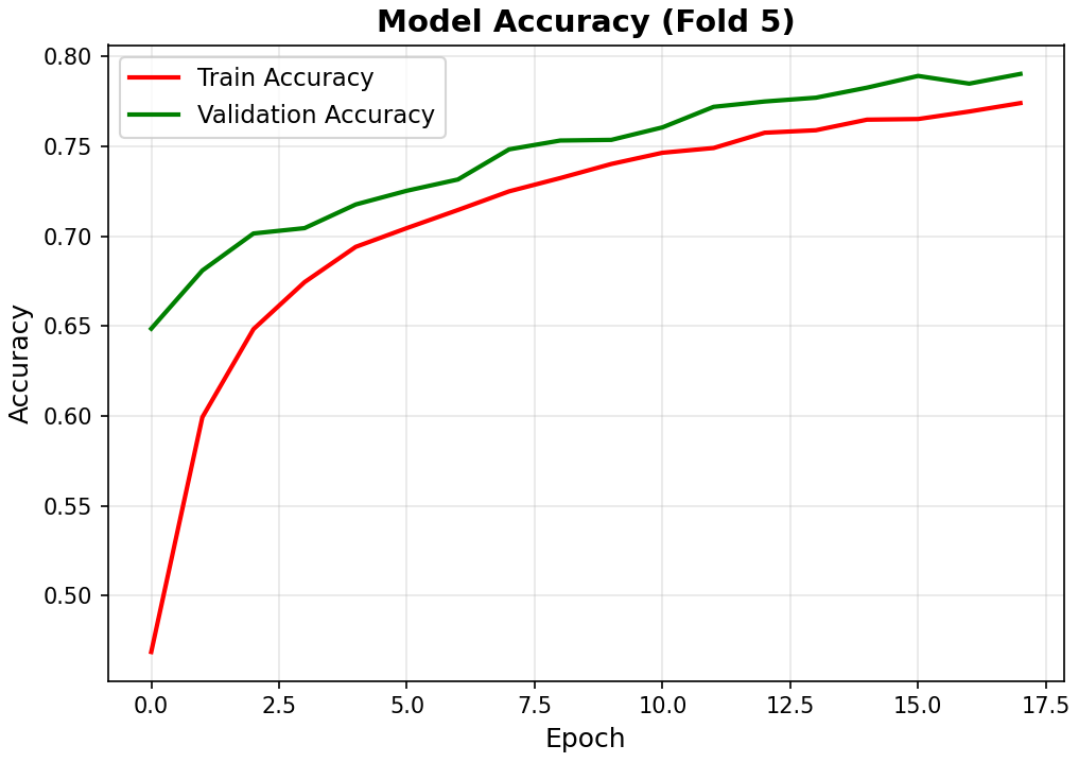

Figure 4.7: Model Accuracy Fold 5

4.4.4 Comparative Analysis

This subsection presents a direct comparison between the baseline model trained without data augmentation and the model trained with CTGAN-based data augmentation. The comparison is conducted at the window level to ensure consistency with the experimental objective and evaluation methodology.

The augmented model demonstrates a clear improvement in overall classification performance compared to the baseline model. Specifically, the window-level classification accuracy increased from 75.6% in the baseline setting to 80.4% after applying data augmentation. This improvement indicates that the additional synthetic samples generated using CTGAN effectively enhanced the model's ability to generalise across different driving behaviour classes.

In terms of class-wise performance, the most significant improvements were observed for Aggressive and Drowsy driving behaviours. For aggressive driving, the F1-score increased substantially due to a marked improvement in recall, indicating that fewer aggressive windows were misclassified after augmentation. Similarly, drowsy driving showed improved recall and F1-score, demonstrating better recognition of under-represented behaviour patterns. These improvements confirm that CTGAN-based augmentation successfully mitigates class imbalance by enriching the training dataset with more diverse minority class samples.

For Normal driving behaviour, the performance remained relatively stable before and after augmentation. This outcome is expected, as the normal driving class was already well represented in the original dataset. Importantly, the improved performance for minority classes did not come at the expense of degraded performance for the majority class, indicating balanced learning behaviour.

The comparative analysis of confusion matrices further supports these findings. After data augmentation, a reduction in misclassification between aggressive and drowsy driving behaviours was observed, along with a clearer diagonal dominance in the confusion matrix. This indicates improved class separation and stronger confidence in model predictions.

Overall, the comparative analysis demonstrates that CTGAN-based data augmentation significantly enhances the robustness and generalisation capability of the proposed hybrid CNN–BiLSTM model. By improving minority class recognition while maintaining stable performance for majority classes, the augmented model provides a more reliable and balanced solution for window-level driving behaviour classification.

4.5 COMPARISON WITH EXISTING METHODS

This subsection presents a comparative study of the proposed hybrid model that uses CTGAN for augmentation with other models in machine learning and deep learning for classifying driving behavior. This comparison will be done on a window-by-window basis to ensure that there is consistency in the experimental design. This is because different models have different applications in classifying driving behavior data.

4.5.1 Linear Support Vector Machine (SVM)

The Linear Support Vector Machine (SVM) has been widely utilized in early research on driving behavior analysis due to its simplicity and efficacy in high-dimensional feature spaces [1]. The objective of the linear SVM is to discover the optimal hyperplane that discriminates classes based on static input features. Nevertheless, the data on driving behavior contain strong non-linear and temporal features that are difficult to represent by a hyperplane. Consequently, the efficacy of the linear SVM is limited in window-based driving behavior classification with overlapping patterns of driving behavior.

The Linear SVM achieved a window-level classification accuracy of 41.49% with a weighted F1-score of 42.62%, indicating limited effectiveness for driving behaviour classification. The model performs relatively well for Aggressive driving, achieving an F1-score of 66.56%, but shows poor performance for Drowsy and Normal driving, with F1-scores of 24.83% and 33.63%, respectively.

The confusion matrix reveals significant misclassification between drowsy and normal behaviours, highlighting the inability of a linear decision boundary to separate complex and overlapping driving patterns. Overall, these results confirm that Linear SVM is unsuitable for window-level driving behaviour analysis compared to deep learning-based approaches that explicitly model temporal dynamics.

Table 4.3: SVM Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.7486	0.5991	0.6656	21958
1 (Drowsy)	0.2548	0.2420	0.2483	20916
2 (Normal)	0.2957	0.3899	0.3363	17078
Overall Accuracy			0.4149	59952
Macro Usage	0.4330	0.4103	0.4167	59952

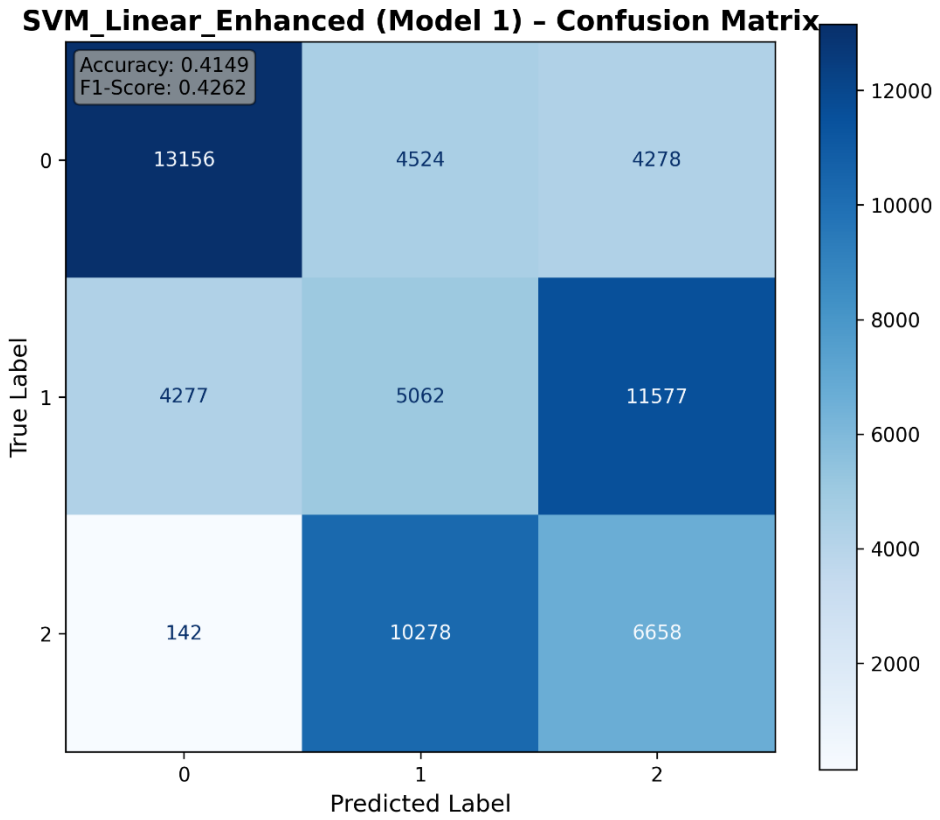

Figure 4.8 SVM Confusion Matrix

4.5.2 Adaptive Boosting (AdaBoost) Classifier

AdaBoost is an ensemble learning algorithm that can improve the accuracy of classification by using a number of weak learners and giving them more weight if they are misclassified during the training process [1]. For driving behavior analysis, AdaBoost can be used to improve accuracy by concentrating on the most difficult samples. However, the method uses statistical features designed by humans and does not consider the temporal dependencies in the driving data. Therefore, the capability of AdaBoost in identifying compound sequential patterns in driving behavior is limited compared to the use of deep learning algorithms.

The AdaBoost classifier achieved a window-level classification accuracy of 29.14% with a weighted F1-score of 31.04%, indicating very weak performance for driving behaviour classification. The model performs relatively well for Aggressive driving in terms of precision (93.44%); however, its recall is low (39.00%), resulting in many aggressive windows being misclassified as other behaviours.

For Drowsy driving, AdaBoost performs particularly poorly, with an F1-score of only 9.38%, indicating a strong bias against this class. Similarly, Normal driving shows low performance with an F1-score of 26.73%. The confusion matrix reveals that the model frequently misclassifies drowsy and normal windows as aggressive driving, suggesting that AdaBoost is heavily influenced by dominant feature patterns and struggles under class imbalance.

Overall, these results demonstrate that AdaBoost is not suitable for window-level driving behaviour classification, as it relies on static features and lacks the ability to model temporal dynamics present in driving data. Compared to deep learning-based approaches, its performance is significantly inferior.

Table 4.4: AdaBoost Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.9344	0.3900	0.5503	21958
1 (Drowsy)	0.1214	0.0764	0.0938	20916
2 (Normal)	0.1943	0.4280	0.2673	17078
Overall Accuracy			0.2914	59952
Macro Usage	0.4167	0.2981	0.3038	59952

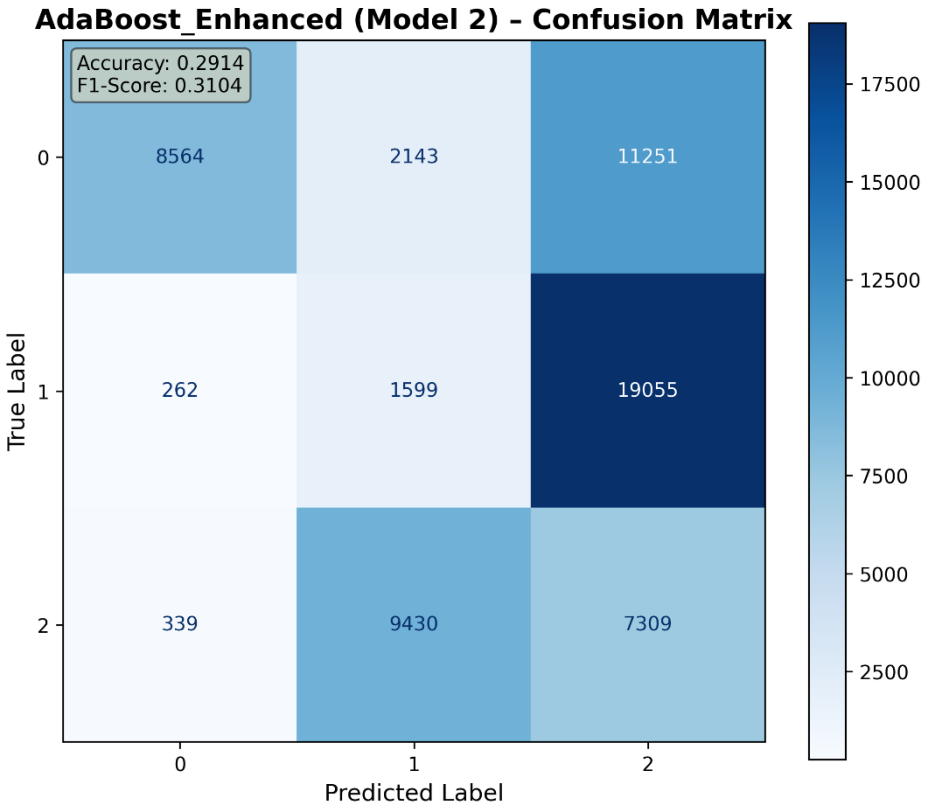

Figure 4.9: AdaBoost Confusion Matrix

4.5.3 Random Forest Classifier

Random Forest is an ensembled classifier. It is known for its robustness to noisy data, ability to model non-linear relationships between variables, and resistance to overfitting due to being an ensembled technique. However, it is observed to be a disadvantage that in a window, data samples are considered to be uncorrelated to each other, paying no heed to the order of sensor readings in a window. Such an issue makes their ability to analyze dynamically changing patterns of driver behaviors weaker than sequence models based on neural networks.

The Random Forest classifier achieved a window-level classification accuracy of 38.90% with a weighted F1-score of 41.77%, indicating limited performance for driving behaviour classification. The model shows moderate performance for Aggressive driving, with an F1-score of 63.65%, but performs poorly for Drowsy and Normal driving, achieving F1-scores of 36.01% and 20.69%, respectively.

The confusion matrix reveals frequent misclassification of drowsy and normal windows as other classes, particularly aggressive driving. Although Random Forest can model non-linear feature interactions, it treats each window independently and does not capture temporal relationships within driving data. Consequently, its performance remains inferior to deep learning-based models that explicitly learn sequential behaviour patterns.

Table 4.5: Random Forest Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.7850	0.5353	0.6365	21958
1 (Drowsy)	0.3892	0.3350	0.3601	20916
2 (Normal)	0.1690	0.2669	0.2069	17078
Overall Accuracy			0.3890	59952
Macro Usage	0.4477	0.3791	0.4012	59952

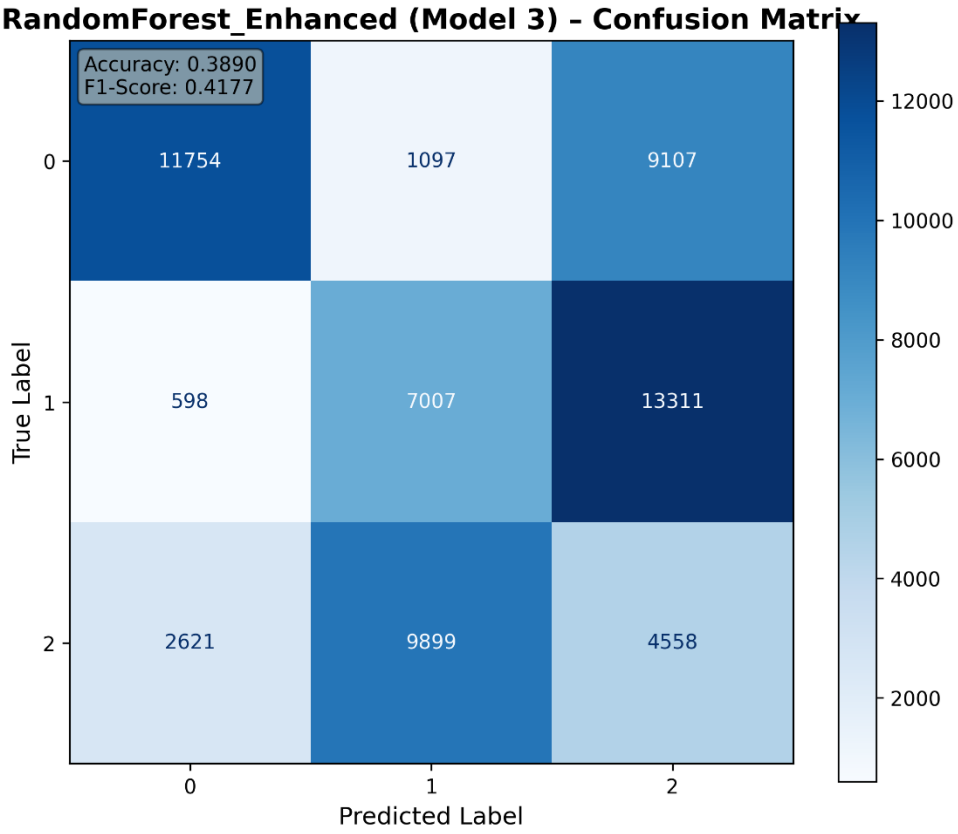

Figure 4.10: Random Forest Confusion Matrix

4.5.4 Gradient Boosting Classifier

Gradient Boosting is a quite effective method in ensemble learning for creating models one by one to minimize misclassifications during classification. It has been applied for classifying driving behavior due to its effective predictive capability for structured data. However, like other ensemble models, it requires a specific feature representation and lacks any temporal representation in its models. As driving behavior varies with time, it may fail during comparison with the recurrent network models that learn by experiencing the sequence.

The best window-level accuracy achieved by the Gradient Boosting classifier was 37.97%, while the weighted F1-score reached 40.86%, demonstrating only modest effectiveness in classifying driving behavior. It performs decently on Aggressive driving, with an F1-score of 58.96%, but for Drowsy and Normal driving, the F1-scores are 35.27% and 24.44%, respectively.

As can be seen in the confusion matrix, drowsy and normal windows are mostly misclassified to be in other behavior classes, especially driving aggressively. Gradient Boosting has the advantage of learning from its previous mistakes, but it relies on fixed feature representation and fails to model temporal dependencies found in driving data, thus making its performance restricted and usually lower than deep learning models designed for time-series analysis.

Table 4.6: Gradient Boosting Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.7858	0.4718	0.5896	21958
1 (Drowsy)	0.3950	0.3186	0.3527	20916
2 (Normal)	0.1920	0.3362	0.2444	17078
Overall Accuracy			0.3797	59952
Macro Usage	0.4576	0.3755	0.3956	59952

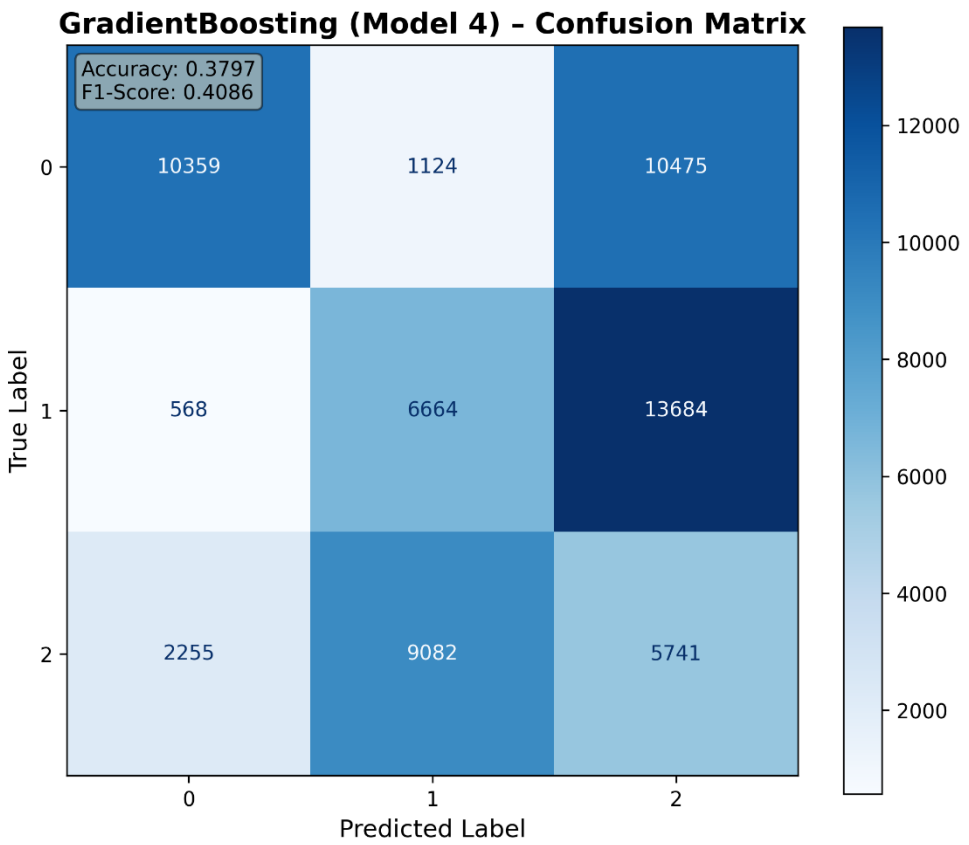

Figure 4.11: Gradient Boosting Confusion Matrix

4.5.5 K-Nearest Neighbours (KNN) Classifier

K-Nearest Neighbours (KNN), being a simplistic, distance-based method, was employed in driving behaviour analysis due to the simplicity of its decision boundary. This algorithm draws a prediction based on its neighbour features in a labelled set. Though KNN can be effective when the class distinction is crisp, it fails when there is noisy data, isn't robust when dealing with feature scaling, or when it's fed an unbalanced class dataset. Furthermore, it is inefficient when dealing with large datasets and lacks an ability to incorporate time features.

The results indicate that the K-Nearest Neighbours classifier achieved a window-level accuracy of 36.04% and a weighted F1-score of 39.18%, which points out a modest success in identifying driving patterns. The classifier performs decently for Aggressive driving with an F1 of about 60.73%, but fares poorly for both Drowsy and Normal driving patterns with respective F1-scores of 29.56% and 23.25%.

On analyzing the confusion matrix, there is a significant overlap between drowsy driving and normal driving, where many samples are being wrongly classified because of similarities between their characteristics. Because KNN is based solely on distances, there are limitations in distinguishing complex patterns in driving behaviors because it doesn't learn from time patterns, hence underperforming in comparison to models that are based on deep learning.

Table 4.7: KNN Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.8407	0.4753	0.6073	21958
1 (Drowsy)	0.3204	0.2743	0.2956	20916
2 (Normal)	0.1833	0.3180	0.2325	17078
Overall Accuracy			0.3604	59952
Macro Usage	0.4481	0.3559	0.3785	59952

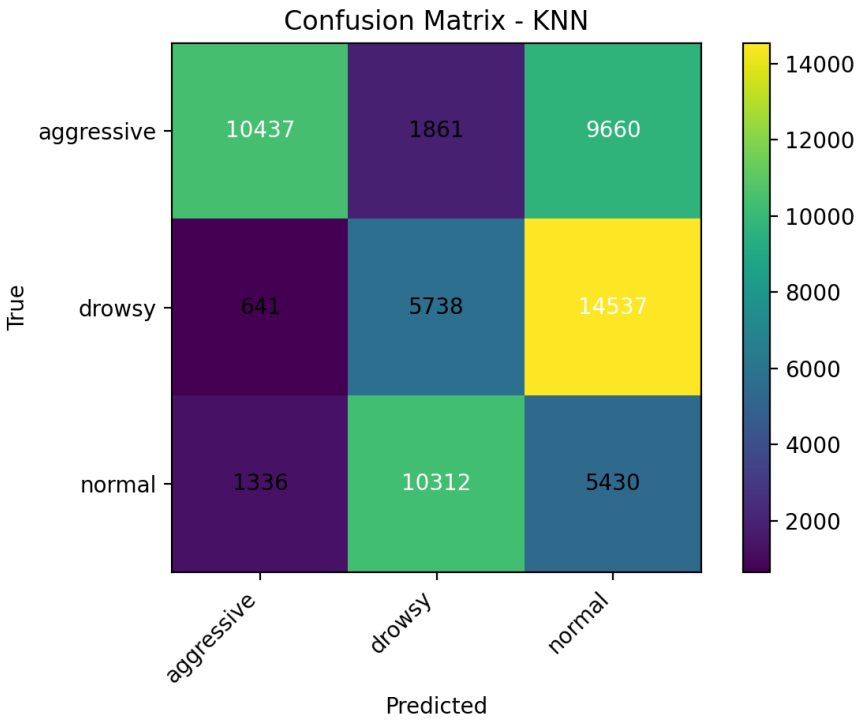

Figure 4.12: KNN Confusion Matrix

4.5.6 CTGAN-Enhanced BiLSTM with Hoeffding Tree Classifier

The CTGAN-enhanced BiLSTM with Hoeffding Tree classifier represents a hybrid framework that combines data augmentation, deep temporal feature extraction, and stream-based classification [12]. In this approach, CTGAN is used to generate synthetic samples for minority classes, while the BiLSTM learns temporal representations from sequential data. The Hoeffding Tree classifier then performs classification on the extracted features. Although this method improves performance compared to traditional machine learning models, the separation between feature extraction and classification limits end-to-end optimisation, resulting in slightly lower performance than fully integrated deep learning architectures.

The CTGAN-enhanced BiLSTM with a Hoeffding tree classifier gave a window-level accuracy of 46.70% with a weighted F1-score of 49.58%, which outperformed conventional machine learning techniques. It particularly excels in identifying Aggressive driving with a precision of 91.79% and F1 score of 72.14%, indicating the efficacy of the use of BiLSTM features in identifying aggressive actions.

However, its performance with Drowsy and Normal drivers remains poor, with only 41.04% and 31.06% accuracy, respectively, on F1 metrics. Also, from the confusion matrix, there is obvious misclassification between these two classes, which indicates that it might be difficult to distinguish between their minute characteristics. Though CTGAN aided in class balancing and BiLSTM accounted for its time-series nature, having a distinct Hoeffding Tree classifier classifies this approach as inferior to that of the former.

Table 4.8:CTGAN-Enhanced BiLSTM with Hoeffding Tree Classifier Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.9179	0.5941	0.7214	4366
1 (Drowsy)	0.4424	0.3827	0.4104	4157
2 (Normal)	0.2513	0.4066	0.3106	3399
Overall Accuracy			0.4670	11922
Macro Usage	0.5372	0.4612	0.4808	11922

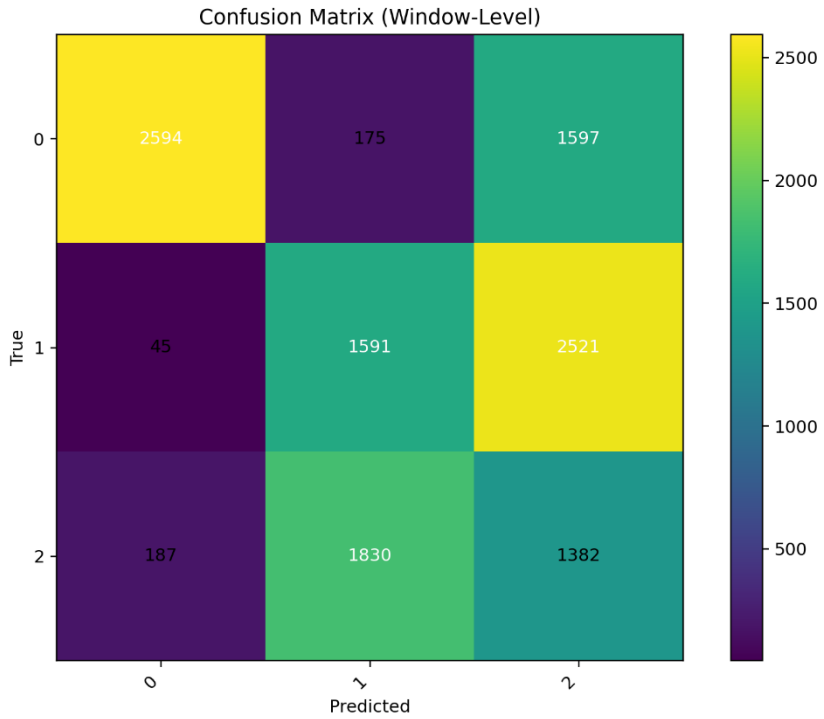

Figure 4.13: CTGAN-Enhanced BiLSTM with Hoeffding Tree Classifier Confusion MATRIX

4.5.7 Long Short-Term Memory (LSTM) Architecture

Long Short-Term Memory (LSTM) Networks, in particular, have become popular in classifying driving behavior because of their ability to extract how things change over time in sequences[14]. They learn patterns from sensor data itself, so they always seem to outperform traditional machine learning classification algorithms. The catch with traditional LSTM Networks, though, is that they flow in one way in time and use only the order of data. They might thus overlook some contextual information in time, where data from after might help infer behavior.

The LSTM model achieved a window-level classification accuracy of 45.32% with a weighted F1-score of 45.89%, showing improved performance compared to traditional machine learning models. The model performs well for Aggressive driving, with an F1-score of 72.17%, indicating that LSTM is effective in learning temporal patterns associated with aggressive behaviour.

However, the performance for Drowsy and Normal driving remains limited, with F1-scores of 43.75% and 14.72%, respectively. The confusion matrix shows considerable misclassification between drowsy and normal behaviours, suggesting that a unidirectional LSTM struggles to fully capture complex contextual information. While LSTM benefits from temporal modelling, its performance is still inferior to the proposed hybrid CNN–BiLSTM model, which incorporates richer feature extraction and bidirectional temporal learning.

Table 4.9: LSTM Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.8305	0.6381	0.7217	21958
1 (Drowsy)	0.3786	0.5183	0.4375	20916
2 (Normal)	0.1607	0.1359	0.1472	17078
Overall Accuracy			0.4532	59952
Macro Usage	0.4566	0.4308	0.4355	59952

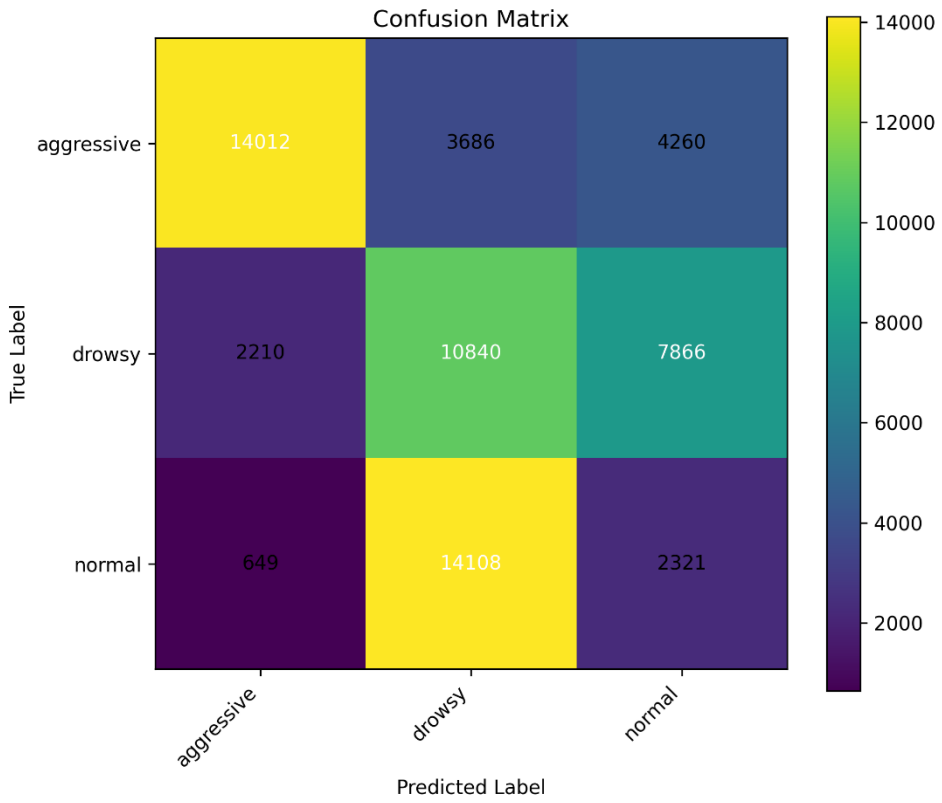

Figure 4.14: LSTM Confusion Matrix

4.5.8 Convolutional Neural Network with Long Short-Term Memory (CNN–LSTM) Architecture

CNN-LSTM architecture combines convolutional layers that extract short-term features of data with an LSTM layer that recognizes long-term relationships within time-series data. This architecture was applied to Driving Behavior classification and normally performs better than traditional machine learning approaches since it learns the appropriate sequence of driving patterns.

However, the CNN-LSTM model has the capability of reading the time information only in one direction. In contrast to this, with the CNN-BiLSTM model, statistical features are introduced along with bidirectional time information. This results in increased accuracy on a window level.

CNN-LSTM configuration achieved a window-level accuracy of 48.31 percent and a weighted F1-score of 46.56 percent, marking a slight improvement upon pure LSTM and conventional machine learning methods. Its strength lies in the area of "Aggressive driving" since it attained a recall of 86.78 percent and an F1-score of 81.04 percent, which indicates that convolutional layers are capable of capturing localized temporal features associated with "Aggressive driving."

On the other hand, the performance on Drowsy and Normal classes is restricted, with F1 scores of 40.48% and 9.72%, respectively. A high number of normal driving windows are confused with drowsy, indicating the model has difficulties distinguishing slight variations while performing a one-directional LSTM-based approach. Despite the superior feature representation obtained from the CNN–LSTM hybrid approach over the LSTM, its performance is still below that of the hybrid proposed approach, which incorporates a bi-directional temporal processing and feature fusion based on the statistical approach and the deep network, respectively.

Table 4.10: CNN LSTM Classification Report

Class	Precision	Recall	F1-score	Support
0 (Aggressive)	0.7601	0.8678	0.8104	4366
1 (Drowsy)	0.4086	0.4010	0.4048	4157
2 (Normal)	0.1064	0.0894	0.0972	3399
Overall Accuracy			0.4831	11922
Macro Usage	0.4250	0.4528	0.4374	11922

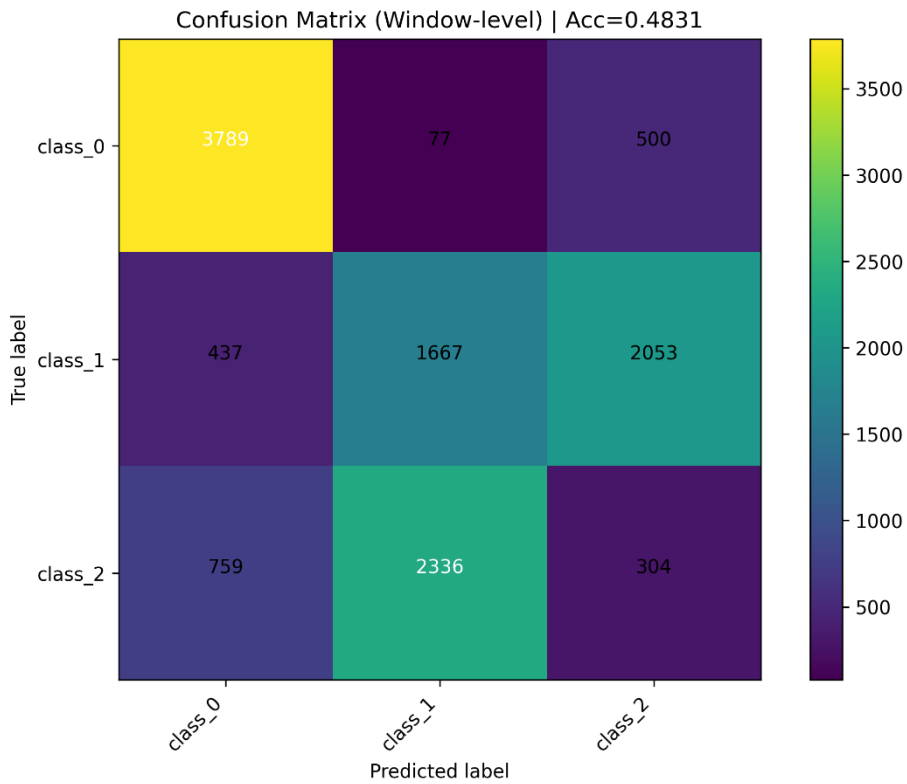

Figure 4.15: CNN LSTM Confusion Matrix

4.5.9 Summary

This chapter presents the empirical evaluation of the proposed driving behavior analysis system for window-level classification. The results indicate that the hybrid deep learning approach successfully captured the temporal driving patterns of identifying aggressive, drowsy, and normal styles. Extensive experiments were conducted, which included baseline tests, data augmentation analysis, training behavior inspection, and comparisons with existing methods.

The baseline hybrid CNN–BiLSTM model ensures good classification, especially for the aggressive driving class, though there is a certain confusion between drowsy and normal states. Application of CTGAN-based data augmentation significantly increases the overall performance with notable gains for the minority behavior classes, which clearly justifies that the generation of synthetic data addresses the problem of class imbalance and improves generalization.

However, SVM, AdaBoost, Random Forest, Gradient Boosting, and KNN, which represent the more classic classifiers, have poor performance in capturing complex temporal dependencies inherent in driving data. Besides, compared to the proposed hybrid architecture, LSTM alone and the CNN–LSTM model achieved limited improvement since they still cannot provide full consideration of subtle variations in behavioral features. In contrast, the proposed hybrid CNN–BiLSTM ensembles local temporal feature extraction, bidirectional sequence modeling, and statistical feature fusion to enable consistent outperformance over others.

In summary, the findings validate the effectiveness of the proposed driving behavior analysis system on window-level classification. The research work further emphasizes the strengths of hybrid architectures and data augmentation for improved robustness. The final chapter concludes with a wrap-up of the study and mentions the possible future research directions.

4.6 KEY FINDINGS

This paper has evaluated the proposal of the driving behavior analysis system with a set of experiments. This paper has compared the baseline hybrid model (CNN-BiLSTM), the model augmented with CT-GAN, along with models prevalent in the literature. The outcome of the paper indicates a number of important things.

Second, integrating CTGAN-synthesized data into the baseline approach improved classification even further, particularly for the minority classes. The synthesized data reduced classification imbalance to a great extent, leading to an increase in recall and F1-measure values for aggressive and drowsy driving classes. The confusion matrix showed improved overall classification with reduced mistakes, proving data augmentation to be an ideal technique for handling classification imbalance in driving data.

Third, when compared with the existing methods, including traditional models (SVM, AdaBoost, RF, GB, KNN Classifier) and other deep models (LSTM, CNN-LSTM, Hybrid models based on CT-GAN), the proposed hybrid CNN-BiLSTM performed the best. Traditional models were poor at modeling the temporal relationships, while some basic deep models were not able to separate the similar states, for example, drowsy and normal state of the driver effectively.

Based on these findings, it can be concluded that all project objectives have been successfully achieved.

The experimental analysis identified clear limitations in existing driving behaviour analysis systems, particularly in handling temporal dependencies and imbalanced data, fulfilling Objective 1. The design and implementation of a hybrid CNN–BiLSTM model successfully enabled accurate classification of normal, aggressive, and drowsy driving behaviours, satisfying Objective 2. The integration of CTGAN-based data augmentation effectively improved model performance for rare driving behaviours, addressing dataset imbalance and achieving Objective 3. Finally, the comprehensive comparison with existing methods confirmed the effectiveness and superiority of the proposed approach, thereby fulfilling Objective 4.

Overall, the key findings demonstrate that the proposed system provides a robust and effective solution for window-level driving behaviour analysis and represents a meaningful improvement over the existing approaches.

Table 4.11 shows the performance of different models in the classification of windows. The conventional machine learning techniques perform poorly, with a score of less than 45%. However, deep learning models perform somewhat better, with a moderate performance of CNN-LSTM in terms of accuracy. The performance of our model, CNN-BiLSTM, significantly surpasses the baselines, and the use of CTGAN for augmentation further enhances the performance.

Table 4.11: Performance Comparison of Existing and Proposed Models at Window Level

No.	Method	Model Type	Window-Level Accuracy (%)
1	SVM (Linear) [1]	Traditional Machine Learning	41.49
2	AdaBoost [1]	Ensemble Machine Learning	29.14
3	Random Forest [1]	Ensemble Machine Learning	38.90
4	Gradient Boosting [1]	Ensemble Machine Learning	37.97
5	K-Nearest Neighbour (KNN) [10]	Distance-based Machine Learning	36.04
6	CTGAN-Enhanced BiLSTM with Hoeffding Tree [12]	Deep Learning + Online Classifier + Data Augmentation	46.70
7	CNN–LSTM [17]	Deep Learning	48.31
8	LSTM Network [14] (Benchmark)	Deep Learning	45.32
9	Proposed CNN–BiLSTM (Baseline)	Hybrid Deep Learning	75.60
10	Proposed CNN–BiLSTM + CTGAN	Hybrid Deep Learning + Data Augmentation	80.40

4.7 LIMITATIONS

Despite the encouraging performance of the proposed driving behaviour analysis system, there are a few shortcomings to be considered. It is purely window-based, leading to classification errors at points in time when the transition from one kind of behavior to another is happening. Despite having a threshold on labeled purity to handle noise, rapid transitions in a window may affect classification accuracy.

The second reason is that this method adopts a fixed window size and step length. The fact is that not all driving patterns progress in the same time scale. Therefore, using one segmentation method might not be suitable for different patterns.

Thirdly, the use of CTGAN in data augmentation can be very beneficial for handling class imbalance issues as well as improving the accuracy of the model by generating synthetic data points; however, the synthetic points may not be representative of the complexity of the driving situation in the real world because some unusual driving conditions might be less represented in the synthetic points.

In addition, the proposed model does not explicitly incorporate contextual driving features such as time of day (day or night), road type (urban, highway, suburban, or residential), vehicle type (car, bus, truck, or motorcycle), traffic conditions (free-flowing or congested), and weather conditions (rainy, sunny, or windy). These contextual factors significantly influence driving behaviour and can cause the same driver to exhibit different driving patterns under different conditions. Since the model relies mainly on sensor-based time-series data, it may not be sufficiently robust to precisely analyse driving behaviour across diverse real-world scenarios.

Finally, the hybrid deep learning architecture requires greater computational resources to train than traditional machine learning. This could possibly prevent this architecture from being directly employed within an embedded system that consumes lower levels of computational resources.

These limitations indicate important directions for further research, which are discussed in the next chapter.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

This research addressed each of its first four objectives, and it accomplished this by examining thoroughly the state of current driving behavior analysis and where it goes wrong. We also discovered that many of the existing state-of-the-art approaches rely on notched indicators and a narrow feature set, which restricts them significantly from being able to detect potential dangers such as aggression or drunk or drowsy driving. This immediately fulfilled Objective 1.

To address this issue, we have developed and used a combination or fusion model that integrates Conv1D and Bidirectional LSTM to take advantage of both models. One reason for this integration is that Conv1D is able to extract short time series behaviors, and Bidirectional LSTM is able to extract long time series behaviors based on time series driving data. This significantly enhances the ability to cope with time series behaviors involved in driving, resulting in Achievement of Objective 2.

Additionally, we addressed the problem of class imbalance since aggressive driving is less common than safe driving. This was done by applying Conditional Tabular Generative Adversarial Networks (CT-GAN), which involved creating synthetic data for the less common classes to improve the detector's performance on aggressive driving and drowsy driving. This addressed Objective 3.

For the purpose of evaluation, we employed a structured training and testing procedure that entailed the use of Stratified Group K-Fold cross-validation and measures related to the accuracy level of the results, precision, recall, F1-measure, and confusion matrix analysis. Evaluation saw the proposed technique perform better and more reliably for the achievement of Objective 4.

In conclusion, the proposed method provides a comprehensive and effective framework for the analysis of driving behavior. The issue of temporal modeling, data imbalance, and robust evaluation can be effectively solved for unsafe driving behavior detection using the proposed method, which can improve the process with the help of data augmentation and hybrid models using deep learning techniques.

5.2 FUTURE WORK

Although the proposed system demonstrates strong performance in driving behaviour analysis using time-series sensor data, several directions can be explored to further enhance its capability and practical applicability. One important area for future work is the incorporation of contextual information alongside sensor data. Factors such as road type (urban, highway, rural), weather conditions, vehicle type, traffic density, and time of day have a significant influence on driving behaviour. Integrating these contextual features with sensor-based inputs could enable the model to better interpret driving actions under different situations and reduce misclassification caused by context-dependent driving patterns.

Future studies may also explore multimodal data fusion, where contextual data from external sources, such as weather services, digital maps, or vehicle type databases, are combined with sensor data in a unified learning framework. This would allow the system to distinguish between risky behaviour and behaviour that is reasonable under specific conditions, such as sudden braking in congested traffic or reduced speed during adverse weather. Incorporating these additional inputs could improve robustness and support more accurate real-world deployment.

In addition, the scope of this project can be extended by evaluating the system in real-time or online learning environments, where the model continuously adapts to new driving data. Future work may also investigate alternative data augmentation strategies or advanced generative models to further improve learning from limited unsafe driving samples.

Finally, the project timeline and development plan for the continuation of this research are illustrated in the Gantt chart presented in Figure 5.1. The chart outlines the planned activities for FYP II, including extended data collection, contextual feature integration, model enhancement, and system validation. This structured timeline provides a clear roadmap for future development and ensures systematic progress toward building a more comprehensive and context-aware driving behaviour analysis system.

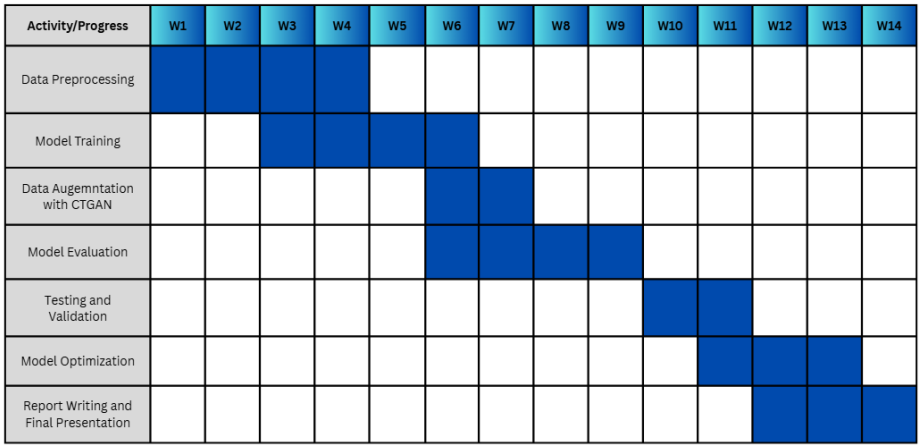

Figure 5.1 Gantt Chart for FYP 2

BIBLIOGRAPHY

[1] R. kumar and A. Jain, "Driving behavior analysis and classification by vehicle OBD data using machine learning," *J. Supercomput.*, vol. 79, no. 16, pp. 18800–18819, Nov. 2023, doi: 10.1007/s11227-023-05364-3.

[2] S. Roussou, T. Garefalakis, E. Michelaraki, T. Brijs, and G. Yannis, "Machine Learning Insights on Driving Behaviour Dynamics among Germany, Belgium, and UK Drivers," *Sustain.*, vol. 16, no. 2, Jan. 2024, doi: 10.3390/su16020518.

[3] S. Roussou *et al.*, "Unfolding the dynamics of driving behavior: a machine learning analysis from Germany and Belgium," *Eur. Transp. Res. Rev.*, vol. 16, no. 1, Dec. 2024, doi: 10.1186/s12544-024-00655-z.

[4] S. Mozaffari, O. Y. Al-Jarrah, M. Dianati, P. Jennings, and A. Mouzakitis, "Deep Learning-Based Vehicle Behavior Prediction for Autonomous Driving Applications: A Review," Jan. 01, 2022, *Institute of Electrical and Electronics Engineers Inc.* doi: 10.1109/TITS.2020.3012034.

[5] M. H. Alkinani, W. Z. Khan, and Q. Arshad, "Detecting Human Driver Inattentive and Aggressive Driving Behavior Using Deep Learning: Recent Advances, Requirements and Open Challenges," *IEEE Access*, vol. 8, pp. 105008–105030, 2020, doi: 10.1109/ACCESS.2020.2999829.

[6] S. Bouhsissin, N. Sael, and F. Benabbou, "Driver Behavior Classification: A Systematic Literature Review," 2023, *Institute of Electrical and Electronics Engineers Inc.* doi: 10.1109/ACCESS.2023.3243865.

[7] Q. Deng and D. Soffker, "A Review of HMM-Based Approaches of Driving Behaviors Recognition and Prediction," Mar. 01, 2022, *Institute of Electrical and Electronics Engineers Inc.* doi: 10.1109/TIV.2021.3065933.

[8] J. Wang *et al.*, "A Survey on Driver Behavior Analysis From In-Vehicle Cameras," Aug. 01, 2022, *Institute of Electrical and Electronics Engineers Inc.* doi: 10.1109/TITS.2021.3126231.

[9] V. Shirole, A. K. Shahade, and P. V. Deshmukh, "A comprehensive review on data-driven driver behaviour scoring in vehicles: technologies, challenges and future directions," Dec. 01, 2025, *Springer Nature*. doi: 10.1007/s44163-025-00244-6.

[10] S. Kumar Singh and A. Sharma, "Revving up insights: machine learning-based classification of OBD II data and driving behavior analysis using g-force metrics," *Bull. Electr. Eng. Informatics*, vol. 14, no. 3, pp. 2188–2197, Jun. 2025, doi: 10.11591/eei.v14i3.9398.

[11] M. A. Uddin *et al.*, "Abnormal Driving Behavior Detection: A Machine and Deep Learning Based Hybrid Model," *Int. J. Intell. Transp. Syst. Res.*, Apr. 2025, doi: 10.1007/s13177-025-00471-2.

[12] S. Arumugam and R. Bhargavi, “Generative AI and Online Learning Based Road Rage and Aggressive Driving Detection,” *Int. J. Intell. Transp. Syst. Res.*, vol. 23, no. 1, pp. 290–314, Apr. 2025, doi: 10.1007/s13177-024-00450-z.

[13] J. Maktoubian, S. N. Tran, A. Shillabeer, M. B. Amin, L. Sambrooks, and R. Khoshkangini, “Empirical Evaluation of Machine Learning Models for Fuel Consumption, Driver Identification, and Behavior Prediction,” *IEEE Trans. Intell. Transp. Syst.*, 2024, doi: 10.1109/TITS.2024.3474745.

[14] A. A. Pande, P. S. Wawage, and Y. D. Deshpande, “Driver Behaviour Classification using Neural Networks,” in *2024 4th Asian Conference on Innovation in Technology, ASIANCON 2024*, Institute of Electrical and Electronics Engineers Inc., 2024. doi: 10.1109/ASIANCON62057.2024.10838053.

[15] I. Sar, A. Routray, and B. Mahanty, “A Review on Existing Technologies for the Identification and Measurement of Abnormal Driving,” *Int. J. Intell. Transp. Syst. Res.*, vol. 21, no. 1, pp. 159–177, Apr. 2023, doi: 10.1007/s13177-023-00343-7.

[16] J. M. Mase, P. Chapman, and G. P. Figueredo, “A Review of Intelligent Systems for Driving Risk Assessment,” *IEEE Trans. Intell. Veh.*, 2023, doi: 10.1109/TIV.2023.3318113.

[17] D. I. Tselentis and E. Papadimitriou, “Driver Profile and Driving Pattern Recognition for Road Safety Assessment: Main Challenges and Future Directions,” *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 83–100, 2023, doi: 10.1109/OJITS.2023.3237177.

[18] K. Ahmad, E. P. Ping, and N. A. Ab Aziz, “Leveraging OBD II Time Series Data for Driver Drowsiness Detection: A Recurrent Neural Networks Approach,” in *6th IEEE International Conference on Artificial Intelligence in Engineering and Technology, IICAIET 2024*, Institute of Electrical and Electronics Engineers Inc., 2024, pp. 518–523. doi: 10.1109/IICAIET62352.2024.10729978.

[19] X. Tao *et al.*, “A multimodal physiological dataset for driving behaviour analysis,” *Sci. Data*, vol. 11, no. 1, Dec. 2024, doi: 10.1038/s41597-024-03222-2.

[20] M. Al-Kahtani, S. Thakur, A. El-Badawy, A. Gupta, M. Elhenawy, and N. Maslekar, "Spatiotemporal Risky Driving Behavior Identification Using Connected Vehicle Trajectory Data for Smart City Applications," *Sustainability*, vol. 15, no. 13, pp. 1–21, Jul. 2023. doi: 10.3390/su151311025.

[21] T. Garefalakis, E. Michelaraki, S. Roussou, C. Katrakazas, T. Brijs, and G. Yannis, “Predicting risky driving behavior with classification algorithms: results from a large-scale field-trial and simulator experiment,” *Eur. Transp. Res. Rev.*, vol. 16, no. 1, Dec. 2024, doi: 10.1186/s12544-024-00691-9.

[22] S. Bouhsissin, N. Sael, F. Benabbou, A. Soultana, and A. Jannani, “SafeSmartDrive: Real-Time Traffic Environment Detection and Driver Behavior Monitoring with Machine and Deep Learning,” *IEEE Access*, 2024, doi: 10.1109/ACCESS.2024.3498596.

[23] E. Romera, L. M. Bergasa, and R. Arroyo, “Need data for driving behavior analysis? Presenting the public UAH-DriveSet,” in *Proceedings of the IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Rio de Janeiro, Brazil, Nov. 2016, pp. 387–392